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套袋及植物生長調節劑對黃晶果(*Pouteria caimito* Radlk.) 果實
品質之影響

Effect of Bagging and Plant Growth Regulators (PGRs) on Fruit
Quality in Abiu (*Pouteria caimito* Radlk.)

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摘要

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論文摘要內容：

本研究利用套袋及噴施激勃素 3(GA_3)、萘乙酸(NAA)、細胞分裂素(Cytokinin)等植物生長調節劑以改進黃晶果之產量及品質。研究包括 3 項試驗，試驗 1 以 10 ppm 之 GA_3 、NAA、Cytokinin 及對照，於 2014 年秋冬兩季噴施於黃晶果樹上。試驗 2 則是黃晶果植株噴施 0 (對照), 10, 50, and 100 ppm 之 GA_3 。試驗 3 則是黃晶果套用白紙袋、黑塑膠袋、白塑膠袋、藍紙袋及對照之套袋處理。植物生長調節素於授粉後 4 周處理，套袋則於開花後 20 天之幼果期處理。試驗 1 結果顯示施用 GA_3 , NAA 及 Cytokinin 可較對照組提高果實長度、直徑、重量、硬度及果肉厚度等果實特性。施用 GA_3 及 NAA 能增加種子大小(長度及重量)。在果實品質方面， GA_3 處理之果實其維生素 C 含量、pH 值及全可溶性固形物(TSS)含量增加。試驗 2 中，噴施 50 ppm 處理者其果實特性種子大小及 pH 值皆較對照與噴施 10 及 100 ppm 者為高。此外，噴施 50 及 100 ppm 之處理，其全可溶性固形物(TSS)含量也較其它處理者為優。10 及 50ppm 的 GA_3 處理能改善果實維生素 C 含量，對照組之維生素 C 含量則是最低。在試驗 1 及 2 中，施用植物生長調節劑者其可滴定酸(TA)含量都會降低。

在試驗 3，以白紙袋套袋處理者，其果實大小、重量、果肉厚度、硬度及維生素 C 含量較其他處理者為高。套袋處理者，其果實可滴定酸(TA)含量會降低。總結言之，施用 50ppm 的 GA_3 能有效地促進黃晶果之生長及果品品質，套袋處理對提高果實特性有顯著之效果。

關鍵字：黃晶果、植物生長調節劑、激勃素 3、萘乙酸、細胞分裂素、套袋、產量、品質

Abstract

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The Contents of Abstract in This Thesis:

Effects of bagging and PGRs on the yield and quality of abiu has been studied for improving fruit quality through bagging and spraying of GA₃, NAA, and cytokinin. Three experiments were conducted in this study. In experiment 1 the 10 ppm of GA₃, NAA, cytokinin and the control were applied to the abiu plants at two seasons (autumn and winter, 2014). In experiment 2 abiu was treated by GA₃ under the concentrations of 0 (control), 10, 50, and 100 ppm. In experiment 3 abiu was bagged with 5 bagging treatments including the control, white paper bag, black plastic bag, white plastic bag, and blue paper bag. PGRs were applied 4 weeks after pollination while bagging fruit was applied in young fruit period, about 20 days after flowering. Results obtained from experiment 1 showed that the application of GA₃, NAA, and cytokinin increased the fruit characteristics (fruit length, diameter, weight, firmness, flesh thickness) compared to the control. Seed size (length and weight) was increased by spraying GA₃ and NAA. With regard to the fruit quality, GA₃ treatment increased vitamin C, pH and TSS. In experiment 2, GA₃ was applied at different concentration levels, fruit characteristics, seed size and pH were higher in 50 ppm compared to control, 10 and 100 ppm. In addition, TSS was better in 50 and 100 ppm compared to the other treatments. 10 and 50 ppm GA₃ treatment improved vitamin C

content. Lowest vitamin C content was found in control treatment. Titratable acidity acid (TA) was decreased in PGRs-treated fruits in the experiment 1 and 2. In Experiment 3, fruit size and weight, flesh thickness, firmness, and vitamin C were highest by bagging with the white paper bag compared to other treatments. The titratable acidity acid (TA) was decreased in treated fruits. It was then concluded that GA₃ at 50 ppm can be an effective way to enhance the fruit growth as well as to improve the fruit quality of abiu. Bagging had a significant effect on the fruit quality characteristics enhancement.

Keywords: Abiu, PGRs, GA₃, NAA, cytokinin, bagging, yield, quality

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Table of Contents

摘要	I
Abstract	III
Acknowledgment	V
List of tables	IX
List of figures	X
1. INTRODUCTION.....	1
2. LITERATURE REVIEW.....	4
2.1 Distribution of Abiu	4
2.2 Description of Abiu (<i>Pouteria caimito</i>).....	5
2.2.1 Botanical classification.....	5
2.2.2 Botanical description	5
2.3 The Fruit.....	6
2.4 Floral Biology and Physiology	7
2.5 Plant Hormone	8
2.5.1 Definition of plant hormones.....	8
2.5.2 The Auxin	10
2.5.3 The Gibberellins.....	14
2.5.4 Cytokinins	17
2.6 Bagging Fruit	19
2.6.1 Effects of bagging on fruit size and weight.....	20
2.6.2 Effects of bagging on fruit colour.....	21
2.6.3 Biotic factors influenced by fruit bagging	22
2.6.4 Effect of bagging on fruit nutrient concentrations.....	23
3. MATERIALS AND METHODS	25
3.1 Plant materials and experimental design.....	25
3.1.1 Experiment I: The effects of plant growth regulators (PGRs) on the productivity and quality of abiu fruit of autumn and winter	25
3.1.2 Experiment II: Influence of Gibberellic acids (GA ₃) at different concentrations on the productivity and quality of abiu fruit.	25

3.1.3 Experiment III: The effects of bagging on the productivity and quality of abiu fruit.	25
3.3 Measurement of physiological parameters	26
3.3.1 Fruit weight and seed weight.....	26
3.3.2 Fruit yield.....	26
3.3.3 Fruit size, seed size and flesh thickness.....	26
3.3.4 Firmness	26
3.4 Measurement of biochemical parameters	27
3.4.1 Total soluble solids	27
3.4.2 Titratable acidity acid	27
3.4.3 Vitamin C content	27
3.4.4 pH value	28
3.4.5 Skin color determination.....	28
3.5 Statistical analysis	28
4. RESULTS AND DISCUSSION	29
4.1. Experiment I: The effect of PGRs on the productivity and quality of Abiu fruit between autumn and winter season.....	29
4.1.1 The effect of PGRs on characteristics of fruit and yield	29
4.1.2 The effect of PGRs on flesh thickness.....	30
4.1.3 The effect of PGRs on seed size	30
4.1.4 The effect of PGRs on Total soluble solids (TSS) and pH value.....	30
4.1.5 The effect of PGRs on Titratable Acidity (TA).....	31
4.1.6 The effect of PGRs on Vitamin C.....	31
4.2. Experiment II: Influence of Gibberellic acids (GA ₃) at different concentrations on the productivity and quality of Abiu fruit.....	41
4.2.1 The effect of GA ₃ on fruit size, flesh thickness and yield.....	41
4.2.2 The effect of GA ₃ on Total soluble solids, Titratable acidity, and pH value.....	42
4.2.3 The effect of GA ₃ on Vitamin C.....	42
4.2.4 The effect of GA ₃ on firmness.....	43

4.2.5 The effect of GA ₃ on weight and length of seeds	43
4.3. Experiment III: The effect of bagging on the productivity and quality of abiu fruit.....	50
4.3.1 The effect of bagging on fruit size, flesh thickness and yield	50
4.3.2 The effect of bagging on fruit color.....	50
4.3.3 The effect of bagging on pH value, vitamin C and Titratable acidity	51
4.3.4 The effect of bagging on firmness	52
4.3.5 The effect of bagging on total soluble solids.....	52
5. CONCLUSIONS	59
6. REFERENCES	61

List of tables

Table 1. The effects of plant growth regulators on fruit characteristics of abiu fruit in autumn 2014.....	33
Table 2. The effects of plant growth regulators on fruit characteristics of abiu fruit in winter 2014.....	33
Table 3. The effects of plant growth regulators on seed size of Abiu fruit in autumn and winter season	35
Table 4. The effects of GA ₃ at different concentrations on fruit size, flesh thickness and yield of abiu fruit.....	44
Table 5. The effects of GA ₃ at different concentrations on TSS, TA and pH of abiu fruit	45
Table 6. Effect of bagging on yield contributing characteristics in abiu fruit.....	53
Table 7. Effect of bagging on fruit color of abiu	54
Table 8. Effect of bag treatments on pH, vitamin C, and titratable acidity in abiu fruit	55

List of figures

Figure 1. Map distribution of Abiu	4
Figure 2. Ripe, yellow and unripe, green Abiu fruits	6
Figure 3. Variation in fruit shape and size	7
Figure 4. The effects of plant growth regulators on flesh thickness in autumn and winter	34
Figure 5. The effects of plant growth regulators on pH value in autumn and winter.....	36
Figure 6. The effects of plant growth regulators on titratable acidity in autumn and winter	37
Figure 7. The effects of plant growth regulators on vitamin C in autumn and winter.....	38
Figure 8. The effects of plant growth regulators on total soluble solids in autumn and winter.....	39
Figure 9. Effect of PGRs on abiu fruit in two seasons	40
Figure 10. Effect of GA ₃ at different concentrations on vitamin C of abiu.....	46
Figure 11. Effect of GA ₃ at different concentrations on firmness of abiu.....	47
Figure 12. Effect of GA ₃ on seed weight and seed length of abiu.....	48
Figure 13. Effect of GA ₃ at different concentrations on fruits and seeds of abiu.....	49
Figure 14. Effect of bagging on fruit firmness of bag	56
Figure 15. Effect of bagging on total soluble solids of abiu fruit	57
Figure 16. Four kinds of bag	58
Figure 17. Effect of bagging on color of abiu fruit.....	58

1. INTRODUCTION

Abiu, or *Pouteria caimito* Radlk.(Syn.*Lucuma caimito* Roem.) is a delicious, nutritious, exotic fruit which is little known. It is a member of the family Sapotaceae, and is native to the Western Amazon basin in South America where it was cultivated by Indian tribes about 10,000 years ago and vernacularly called "abi", "caimo", or "cauje". Clement (1989) proposed that the centre of diversity of abiu is located largely between the confluence of the Napo and Ica rivers with the Amazon. Abiu has two infraspecific taxa or landraces defined by fruit shape and size: an ovoid, small fruit which is more primitive and more widely distributed than a spherical, larger fruit which is more advanced and more modified in shape but more restricted in distribution (Calzavara, 1970).

Also, the more primitive smaller fruit has moderate rind latex while the more advanced larger, spherical fruit has lower rind latex. A landrace is a primitive cultivar of a crop plant and is genetically very heterogenous, containing numerous alleles that contribute to the survival of the crop in its natural ecosystem. Thus, landraces are rich sources from which plant breeders can selectively reintroduce alleles into highly bred cultivars. Besides countries in South America, the fruit is also grown in Florida and Northern Australia. It grows well, flowers and fruits prolifically around Darwin and northern Queensland. It bears an attractive, bright yellow fruit with a sweet, caramel-flavored, creamy-white pulp. Abiu has good potential for commercial development with more promotion and publicity.

It has been well documented that the size and quality of the fruits can be affected by certain horticultural cultural practices, such as the application of plant growth hormones (Guardiola, 1992). Gibberellic acid (GA_3) has been shown to increase fruit set and growth in clementine orange, spraying GA_3 increased the fruit size and firmness in cherry fruits (Van Rensburg *et al.*, 1996). In addition to this, El-Sese (2005) working on Balady mandarin trees reported that treatment with GA_3 increased the yield of fruits. GA_3 increased fruit firmness, soluble solids and fruit weight (Basak *et al.*, 1998).

NAA is frequently used for post-bloom thinning of ‘Delicious’ apple fruit. Although NAA is effective in inducing fruit abscission, the persisting fruit often fail to size adequately. This negative effect of NAA on fruit size was first reported by Greene (1943), and remains an important limitation in the use of NAA by the apple industry (Unrath, 1981). The recommended practice for thinning spur-type ‘Delicious’ in Michigan is 10 to 15 mg 1-l NAA applied in a high volume spray at 5 to 15 days after petal fall (DAPF) (Hull, 1993) or 10 to 12 mm king fruit diameter (RFD). Timing of NAA application for maximum abscission is better determined by fruit size than by DAPF or days after full bloom (DAFB) (Donoho, 1968; Leuty, 1973).

Cytokinin increased fruit size and provide feedback flowering (Greene *et al.*, 1990; Ferree, 1996). Link (2000) has noted that the use of Cytokinin increased the fruit size without affecting the spacing. The combined use of compounds for the thinning may result in more intensive thinning compared to a single application of plant regulators (Wertheim, 1997). Good results in terms of increasing the weight and diameter of fruits of “Golden Delicious Clone B” were achieved using BA (Vračević *et al.*, 2008). With the cultivar “Summerred” applied Cytokinin led to an increase in fruit size (Stopar and Lokar, 2003) and inhibited the development of the fruit in the cultivar “Delicious Red Chief” (Bukovac *et al.*, 1994) and the cultivar “Fuji” (Stopar and Tojnko, 2005).

During growth and maturation, fruit are susceptible to insect pests, birds, diseases and mechanical damage, which reduce their quality and commercial value. Bagging, a physical protection technique commonly applied to many fruits, not only improves fruit visual quality by promoting fruit coloration and reducing the incidence of fruit cracking and russet but can also change the microenvironment of fruit development, which has multiple effects on internal fruit quality (Xu *et al.*, 2010). The practice of pre-harvest bagging has been used extensively in several fruit crops to improve skin colour and to reduce mechanical damage as well as sunburn of the skin (Noro *et al.*, 1989; Tyas *et al.*, 1998; Amarante *et al.*, 2002b). Pre-harvest bagging also reduces

agrochemical residues in the fruit and improves insect, disease, and bird damage control (Kitagawa *et al.*, 1992). Several authors have reported contradictory results for the effects of pre-harvest bagging on fruit size, maturity, skin colour and flesh mineral content at harvest for various fruits. This may reflect differences in the type of bag used, the fruit stage when it was bagged, the duration of fruit exposure to natural light after bag removal (before harvesting), and/or fruit-and cultivar-specific responses (Fan and Mattheis, 1998). Fruit bagging also affects internal qualities, such as fruit sweetness and acidity. Most studies indicated that fruit bagging decreased soluble solids content in Asian pear (Huang *et al.*, 2007).

There are little researches has been done to study abiu fruit quality parameters in relation to horticultural practices such as bagging, GA₃, NAA and cytokinin. This study is aimed to develop better understanding of yield and fruit quality parameters in relation to specific horticultural practices (bagging), and plant growth regulators application at different concentrations on yield and quality of fruit.

2. LITERATURE REVIEW

2.1 Distribution of Abiu

The abiu, *Pouteria caimito*, is commonly considered as native to the headwaters of the Amazon (Figure 1). It grows wild in the lower eastern part of the Andes from southwestern Venezuela to Peru. It also grows around Tingo Maria and Iquitos. Peru and it will commonly be found in the Province of Guayas in Ecuador, where it is sold in the market. The Abiu was cultivated by Amerindians and it became widespread in the Amazon, however the origin of the fruits distribution outside the Amazon is uncertain. In the Amazon basin, it is found to grow heavily in the Northern Brazilian State of Para but is also found sparsely in collections from the Atlantic rainforest near Rio de Janeiro and Bahia (Mortan, 1987). It can also be found in Colombia in areas such as the regions of Caqueta. Meta and Vaupes and it is very plentiful in Amazonas, Venezuela. It has also been growing for a very long time in Trinidad.

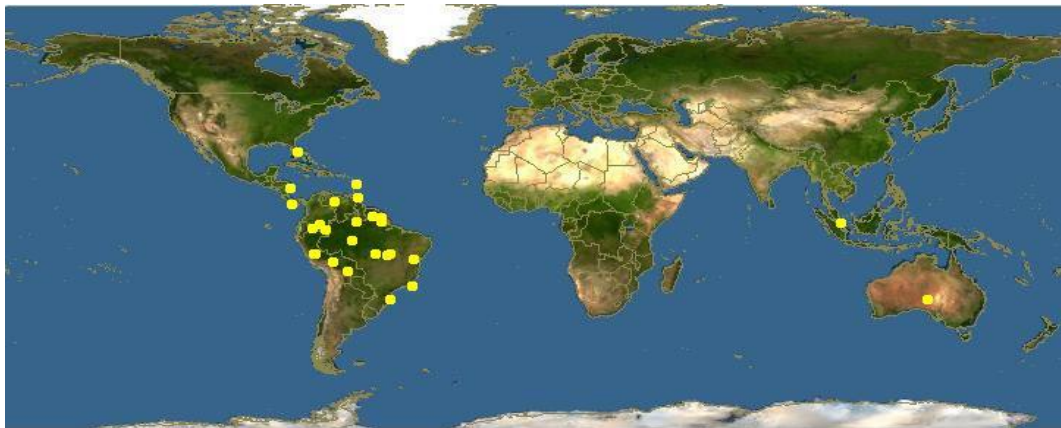


Figure 1. Map distribution of Abiu (Mortan, 1987)

The abiu grows best in areas that have a year-round moist and a warm climate. It can now be found throughout most of the Amazon. It is a common dooryard tree in the backyards and streets in the city of many Brazilian towns, but it is not usually grown commercially. The Abiu habitats are nearly all tropical. It will thrive in a place that has a year round warm and moist climate, although it has been known to grow well in Rio de Janeiro, which is a

somewhat cooler climate. In Peru it cannot grow above 2,000 feet (610 m) in elevation, but in Colombia it has been found growing up to an elevation of 6,000 feet (1,800 m).

2.2 Description of Abiu (*Pouteria caimito*)

2.2.1 Botanical classification

Division:	Spermatophyta
Class:	Angiospermae
Subclass:	Di cotyledoneae
Super order:	Caryophyllidae
Order:	Ebenales
Family:	Sapotaceae
Genus:	<i>Pouteria</i>
Species:	<i>Caimito</i>

Synonyms

Achras caimito Benth

Guapeda caimito Pierre

Labatia caimito Mart

Lucuma caimito Roem & Sch

Pouteria leu-cophaea Baehni

2.2.2 Botanical description

This small to medium sized evergreen tree can grow to over 30 feet in height under optimal conditions but usually is kept low to facilitate harvesting. The rough bark is grey to brown and exudes white latex when cut. The glossy leaves are pointed at both ends and tend to cluster at the end of the branch. Flowers occur along branches between the branch tip and trunk, always in leaf axils. Trees are relatively quick to produce fruit, sometimes in as few as two years or as much as 10 years. Environmental factors also play a part in the time it takes to fruit. Fruit development takes up to 3 months, depending on rainfall or irrigation.

When immature, all parts of the fruit contain sticky white latex, and the cut surface browns rapidly. As the fruit matures, the latex disappears from the pulp and remains only in the peel. The smooth, bright yellow-skinned fruit can be oblong or round and differs in size from 2.5 inches in length to as large 7 inches. The translucent flesh becomes jelly-like, with a pleasant caramel-flavored pulp. The leathery skin can easily be bruised (Popenoe and Wilson, 1920).

2.3 The Fruit

The fruit is a berry, green when immature and turns a bright yellow when mature (Figure 2), elliptical to round, spherical or compressed, often pointed at the stigma end (Figure 3), 60 – 100 mm diameter and weights 100 – 600gm but often up to 1500gm (Lim and Ramsay, 1991).



Figure 2. Ripe, yellow and unripe, green Abiu fruits (Lim and Ramsay, 1991)

The leathery pericarp (rind) is 5mm thick and contains latex which coagulates on exposure to the air when unripe. This latex can irritate the lips or taste buds. The edible pulp is soft, gelatinous, translucent and white. Each fruit has 1 - 4 large seeds, ellipsoidal, 14mm diameter and 37.8mm length, distinct hilum along the lateral and distinct micropyle at one end. The seeds germinate readily when fresh with germination occurring from 2 - 10 days. The seed can germinate while still in the fruit. The seeds retained their

viability for 1 - 2 days in dry conditions indicating that they may be recalcitrant in nature and difficult to store for long periods without losing their viability.

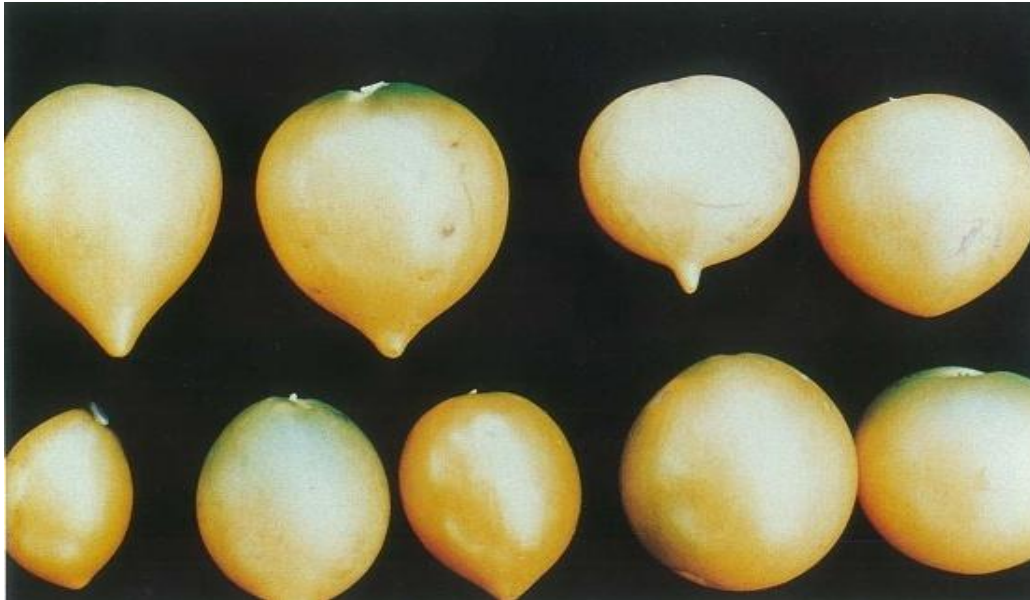


Figure 3. Variation in fruit shape and size (Lim and Ramsay, 1991)

2.4 Floral Biology and Physiology

Abiu is usually an allogamous (cross-pollinated) species but occasionally can be autogamous i.e. self-pollinated (Clement, 1989). The number of fl

owers which open in any one day during the first week of flowering is about 10 - 15% of the total in a flowering terminal. Usually the larger flower buds in a cluster open first, and the flowers along the proximal end of the flowering branch tend to open earlier than those clusters nearer the tip of the branch, in subsequent weeks more of the flower buds open irrespective of their position on the flowering terminal. A day or two before opening, the sepals enlarge first, exposing a tiny slit of 0.5mm at the top. The style carrying the stigma protrudes through the slit and extends 2 – 3mm above the slit, by which time the petals enlarge and elongate and the aperture of the slit reaches 2.5mm. The petals never open fully as in other flowers. Once out of the corolla tube the stigma becomes receptive. The filament also elongates slightly above the ovary hairs but never above the mouth of the corolla tube as

does the style. Another dehiscence occurs before the petal opens and before the stigma becomes receptive, thus favoring cross-pollination. Pollination has been reported to be dependent on high relative humidity and insects like flies, ants and various kinds of butter flies which are yet to be identified and studied. Nectar has been detected in the flowers (Moncur, 1988). Generally, flowers open from 07h30 to 12h00 in the morning, and from 08h30 to 10h30 insect activities is high around the flowers, tapering off after 1100. Thirty to forty hours after flower opening, the petals start to wither, and turn brown. In case of successful pollination and fertilization, the ovary develops into a young fruit about 2 – 3mm in diameter and is covered in the fine hairs. The hairs are shed when the fruit reaches about 6 – 8mm diameter and the stigma begins to shrivel and dry up.

Generally, the fruit is available most of the year in Australia, especially from December to June. In Queensland, three crops can be obtained per year and flowering overlaps with fruit set which is reported to be most abundant during the warmer months. In the NT, there appear to be two crops with the earlier crop in June – July resulting from a flowering in February – March and another crop in November – December from a flowering in July – August. In its native Iquitos, Peru, flowering occurs in October and the crop is harvested in April the following year (Heinz, 1983). Abiu appears not to have any photoperiod requirements for flowering and requires no specific dry duration to trigger flowering unlike many tropical fruit crops. Flowering and fruiting of Abiu are quite prolific in the NT and pose no constraint; Abiu fruit takes over three months to develop to maturity. At three years, a yield of 10kg/tree can be obtained in Queensland and up to 200kg/tree for older trees.

2.5 Plant Hormone

2.5.1 Definition of plant hormones

Plant hormones (also known as phytohormones) are a group of naturally occurring, organic substances which influence physiological processes at low concentrations. The processes influenced consist mainly of growth,

differentiation and development, though other processes, such as stomata movement, may also be affected.

The word hormone is derived from Greek, meaning set in motion. Plant hormones affect gene expression and transcription levels, cellular division, and growth (Srivastava, 2002). They are used to regulate the growth of cultivated plants, weeds, and in vitro-grown plants and plant cells; these manmade compounds are called Plant Growth Regulators or PGRs for short. Early in the study of plant hormones, “phytohormones” was the commonly used term, but its use is less widely applied now.

Plant hormones are signal molecules produced within the plant, and occur in extremely low concentrations. Hormones regulate cellular processes in targeted cells locally and, moved to other locations, in other functional part of the plant. Hormones also determine the formation of flowers, stems, leaves, the shedding of leaves, and the development and ripening of fruit. Plant hormones shape the plant, affecting seed growth, time of flowering, the sex of flowers, senescence of leaves, and fruit. They affect which tissues grow upward and which grow downward, leaf formation and stem growth, fruit development and ripening, plant longevity, and even plant death. Hormones are vital to plant growth, and lacking them, plants would be mostly a mass of undifferentiated cells. So they are also known as growth factors or growth hormones.

Plant hormones are not nutrients, but chemicals that in small amounts promote and influence the growth, (Opik *et al.*, 2005) development, and differentiation of cells and tissues. Hormones are transported within the plant by utilizing four types of movements. For localized movement, cytoplasmic streaming within cells and slow diffusion of ions and molecules between cells are utilized. Vascular tissues are used to move hormones from one part of the plant to another; these include sieve tubes or phloem that move sugars from the leaves to the roots and flowers, and xylem that moves water and mineral solutes from the roots to the foliage.

Plants need hormones at very specific times during plant growth and at

specific locations. They also need to disengage the effects that hormones have when they are no longer needed. Plants use different pathways to regulate internal hormone quantities and moderate their effects; they can regulate the amount of chemicals used to biosynthesize hormones. They can store them in cells, inactivate them, or cannibalise already-formed hormones by conjugating them with carbohydrates, amino acids, or peptides. Plants can also break down hormones chemically, effectively destroying them. Plant hormones frequently regulate the concentrations of other plant hormones (Swarup *et al.*, 2007). Plants also move hormones around the plant diluting their concentrations.

2.5.2 The Auxin

The term auxin is derived from the Greek word 'Auxein' which means to grow. Compounds are generally considered as auxins if they can induce cell elongation in stems and otherwise resemble Indole Acetic Acid (the first auxin isolated) in physiological activity. Auxins usually affect other processes in addition to cell elongation of stem cells but this characteristic is considered critical of all auxins and thus "helps" define the hormone (Arteca, 1996; Mauseth, 1991; Raven *et al.*, 1992; Salisbury and Ross, 1992).

2.5.2.1 History of Auxins

In 1881, Charles Darwin and his son Francis performed experiments on coleoptiles, the sheaths enclosing young leaves in germinating grass seedlings. The experiment exposed the coleoptiles to light from a unidirectional source and observed that they bend towards the light (Hohm *et al.*, 2013). By covering various parts of the coleoptiles with a light impermeable opaque cap, the Darwins discovered that light is detected by the coleoptiles tip, but that bending occurs in the hypocotyls. However, the seedlings showed no signs of development towards light if the tip was covered with an opaque cap, or if the tip was removed. The Darwins concluded that the tip of the coleoptiles was responsible for sensing light, and

proposed that a messenger is transmitted in a downward direction from the tip of the coleoptiles, causing it to bend (Whippo and Hangarter, 2006).

In 1885, Salkowski discovered indole-3-acetic acid (IAA) in fermentation media. The separation of the same product from plant tissues was not found in plant tissues for almost 50 years. IAA is the major auxin involved in many of the physiological processes in plants (Arteca, 1996). Fitting (1907) studied the effect of making incisions on either the light or dark side of the plant. His results aimed to understand if translocation of the signal occurred on a particular side of the plant but his results were inconclusive because the signal was capable of crossing or going around the incision.

In 1913 a Danish scientist named Peter Boysen-Jensen demonstrated that the signal was not transfixed but mobile. He separated the tip from the remainder of the coleoptiles by a cube of gelatin which prevented cellular contact but, allowed chemicals to pass through. The seedlings responded normally bending towards the light. However when the tip was separated by an impermeable substance, there was no curvature of the stem (Boysen-Jensen, 1913).

In 1926, the Dutch botanist Frits Warmolt Went showed that a chemical messenger diffuses from coleoptiles tips. Went's experiment identified how a growth promoting chemical causes a coleoptiles to grow towards light. Went cut the tips of the coleoptiles and placed them in the dark, putting a few tips on agar blocks that he predicted would absorb the growth-promoting chemical. On control coleoptiles, he placed a block that lacked the chemical. On others, he placed blocks containing the chemical, either center on top of the coleoptiles to distribute the chemical evenly or offset to increase the concentration on one side (Whippo and Hangarter, 2006). When the growth promoting chemical was distributed evenly the coleoptiles grew straight. If the chemical was distributed unevenly, the coleoptiles curved away from the side with the cube, as if growing towards light, even though it was grown in the dark. Went later proposed that the messenger substance is a growth-

promoting hormone, which he named Auxin, that becomes asymmetrically distributed in the bending region.

2.5.2.2 Biosynthesis and Metabolism of Auxin

IAA is chemically similar to the amino acid tryptophan which is generally accepted to be the molecule from which IAA is derived. Three mechanisms have been suggested to explain this conversion:

1- Tryptophan is converted to indolepyruvic acid through a transamination reaction. Indolepyruvic acid is then converted to indoleacetaldehyde by a decarboxylation reaction. The final step involves oxidation of indoleacetaldehyde resulting in indoleacetic acid.

2- Tryptophan undergoes decarboxylation resulting in tryptamine. Tryptamine is then oxidized and deaminated to produce indoleacetaldehyde. This molecule is further oxidized to produce indoleacetic acid.

3- As recently as 1991, this third mechanism has evolved. IAA can be produced via a tryptophan-independent mechanism. This mechanism is poorly understood, but has been proven using tryptophan mutants. Other experiments have shown that, in some plants, this mechanism is actually the preferred mechanism of IAA biosynthesis.

The enzymes responsible for the biosynthesis of IAA are most active in young tissues such as shoot apical meristems and growing leaves and fruits. The same tissues are the locations where the highest concentrations of IAA are found. One way plants can control the amount of IAA present in tissues at a particular time by controlling the biosynthesis of the hormone. Another control mechanism involves the production of conjugates which are, in simple terms; molecules resemble to the hormone but are inactive. The formation of conjugates may be a mechanism of storing and transporting the active hormone. Conjugates can be formed from IAA via hydrolase enzymes. Conjugates can be rapidly activated by environmental stimuli signaling a quick hormonal response. Degradation of auxin is the final method of controlling auxin levels. This process also has two proposed mechanisms outlined below:

1- The oxidation of IAA by oxygen resulting in the loss of the carboxyl group and 3- methyleneoxindole as the major breakdown product. IAA oxidase is the enzyme which catalyzes this activity. Conjugates of IAA and synthetic auxins such as 2,4-D cannot be destroyed by this activity .

2- C-2 of the heterocyclic ring may be oxidized resulting in oxindole-3-acetic acid. C-3 may be oxidized in addition to C-2 resulting in dioxindole-3-acetic acid.

The mechanisms by which biosynthesis and degradation of auxin molecules occur are important to future agricultural applications. Information regarding auxin metabolism would most likely lead to genetic and chemical manipulation of endogenous hormone levels resulting in desirable growth and differentiation of important crop species. Ultimately, the possibility exists to regulate plant growth without the use of hazardous herbicides and fertilizers (Davies, 1995; Salisbury and Ross, 1992).

2.5.2.3 Naphthaleneacetic acid (NAA)

NAA is a synthetic plant hormone in the auxin family and is an ingredient in many commercial plant rooting horticultural products; it is a rooting agent and used for the vegetative propagation of plants from stem and leaf cutting. It is also used for plant tissue culture (Morikawa and Takahashi, 2004).

The hormone NAA does not occur naturally, and, like all auxins, is toxic to plants at high concentrations. In the United States, under the Federal Insecticide, Fungicide, and Rodenticide Act (FIFRA), products containing NAA require registration with the Environmental Protection Agency (EPA) as pesticides.

NAA is widely used in agriculture for various purposes. It is considered to be only slightly toxic but when at higher concentrations it can be toxic to animals. This was shown when tested on rats via oral ingestion at 1000 – 5900 mg/kg (Tomlin, 2006). NAA has been shown to greatly increase cellulose fiber formation in plants when paired with another phytohormone called Gibberellic Acid. Because it is in the auxin family it has also been

understood to prevent premature dropping and thinning of fruits from stems. It is applied after blossom fertilization. Increased amounts of it can actually have negative effects however, and cause growth inhibition to the development of plant crops. It has been used on many different crops including apples, olives, oranges, potatoes, and various other hanging fruits. In order for it to obtain its desired effects it must be applied in concentrations ranging from 20 – 100ug/mL (Navalón *et al.*, 1997). NAA present in the environment undergoes oxidation reactions with hydroxyl radicals and sulphate radicals. Radical reactions of NAA were studied by using pulse radiolysis technique. Hydroxyl adduct radical was formed as the intermediate during the reaction of hydroxyl radical with NAA. The intermediate Naphthyl methyl radical was formed during the reaction of sulphate radical anion with NAA (Naduvilpurakkal *et al.*, 2014).

2.5.3 The Gibberellins

Unlike auxins, which are classified on the basis of function, gibberellins are classified on the basis of structure as well as function. All gibberellins are derived from the ent-gibberellane skeleton. The structures of this skeleton derivative along with the structure of a few of the active gibberellins are shown above. All gibberellins are acidic compounds and are therefore also called gibberellic acids (GA) with a different subscript to distinguish between them. GA₃ has historically been called gibberellic acid but the term is also often used in describing all gibberellins. GAs are widespread and so far ubiquitous in flowering (angiosperms) and non-flowering (gymnosperms) plants as well as ferns. They have also been isolated from lower plants such as mosses and algae, at least two fungal species and most recently from two bacterial species. There have been over 90 GAs isolated, all of which are most likely not essential to the plant. Instead, these forms are probably inactive precursors or breakdown products of active gibberellins (Arteca, 1996; Mauseth, 1991; Raven, 1992; Salisbury and Ross, 1992).

2.5.3.1 History of Gibberellins

Japanese farmers first observed the phenomenon of abnormal elongation in certain rice plants early in the season. These plants often became unhealthy and sterile. The agent of the disease bakanae was deduced as being a fungal pathogen of the genus *Fusarium* (Hori, 1898). Kurusawa (1926) discovered that the disease was caused by a substance secreted by the fungal species *Gibberella fujikuroi* resulting to controversy over the true pathogen (Kurusawa, 1926). Wollenweber (1931) stated that the fungus *Fusarium moniliforme* Sheld., which was the asexual or imperfect stage of the ascomycete *Gibberella fujikuroi* (Saw.) Wr. was the culprit for the disease bakanae (Wollenweber, 1931). Yabuta (1935) isolated the compound from *Gibberella fujikuroi* and called it gibberellin A. This compound was found to stimulate growth when applied to rice roots (Yabuta, 1935). Due to Second World War (WWII), much of the work on gibberellins was put on hold and the Western civilizations did not have access to these findings (Arteca, 1996). A new compound from *G. fujikuroi* was discovered in Britain. This compound was named gibberellic acid (Brian et al., 1954). In 1955, a similar compound was also isolated by American scientists from *G. fujikuroi* and which they called gibberellin X (Stodola et al., 1955). Around the same period, Japanese scientists discovered that gibberellin was actually made up of three compounds which they called GA₁, GA₂, and GA₃. Gibberellin X, GA₃, and gibberellic acid are all the same compound. The latter two were accepted in describing the compound and are synonymous terms today (Takahashi et al., 1955). Radley (1956) described some compounds similar to gibberellic acid in plants (Salisbury and Ross, 1992). Takahashi (1957) isolated another compound from *G. fujikuroi* which he called GA₄. He showed that GA₁ was identical to what Stodola and his associates were calling gibberellin A (Takahashi, 1957). MacMillan and Suter (1958) isolated and identified GA₁ from plants in the same year, West and Murashige also identified GA₁ in higher plants (Salisbury and Ross, 1992). MacMillan and Takahashi (1968) proposed that Gibberellins were

assigned numbers in order to reduce confusion between the compounds (Takahashi *et al.*, 1991). This idea proved to be a good one since the procedure is currently used and it is helpful in reducing confusion between the over 90 gibberellins known.

2.5.3.2 Gibberellin Biosynthesis and Metabolism

Gibberellins are usually produced from the methylerythritol phosphate (MEP) pathway in higher plants. In this pathway, bioactive GA is produced from trans-geranylgeranyl diphosphate (GGDP) (Hedden and Thomas, 2012). In the MEP pathway, three classes of enzymes are used to yield GA from GGDP:

1. Terpene synthases (TPSs)
2. Cytochrome P450 mono oxygenases (P450s)
3. 2-oxyglutarate-dependent dioxygenases (2ODDs)

There are 8 steps in the methylerythritol phosphate pathway:

1. GGDP is converted to ent-copalyl diphosphate (ent-CPD) by ent-copalyl diphosphate synthase
2. ent-CPD is converted to ent-kaurene by ent-kaurene synthase
3. ent-kaurene is converted to ent-kaurenol by ent-kaurene oxidase (KO)
4. ent-kaurenol is converted to ent-kaurenal by KO
5. ent-kaurenal is converted to ent-kaurenoic acid by KO
6. ent-kaurenoic acid is converted to ent-7a-hydroxykaurenoic acid by ent-kaurene acid oxidase (KAO)
7. ent-7a-hydroxykaurenoic acid is converted to GA12-aldehyde by KAO
8. GA12-aldehyde is converted to GA12 by KAO.

GA12 is processed to the bioactive GA4 by oxidations on C-20 and C-3, which is accomplished by 2 soluble ODDs: GA 20-oxidase and GA 3-oxidase (Yamaguchi, 2008).

Most bioactive GAs is located in actively growing organs on plants (Hedden and Thomas, 2012). Both GA₂ ox and GA₃ ox genes (genes coding for GA 20-oxidase and GA 3-oxidase) and the SLENDER 1 gene (a GA signal transduction gene) are found in growing organs on rice, which suggests

bioactive GA synthesis occurs at their site of action in growing organs in plants (). During flower development, the tapetum of anthers is believed to be a primary site of GA biosynthesis (Kaneko *et al.*, 2003; Itoh *et al.*, 1999).

Arabidopsis, a plant, and *Gibberella fujikuroi*, a fungus, possess different GA pathways and enzymes (Yamaguchi, 2008). P450s in fungi perform functions analogous to the functions of KAOs in plants (Tudzynski *et al.*, 2001). The function of CPS and KS in plants is performed by a single enzyme, CPS/KS, in fungi (Kawaide *et al.*, 1997; Toyomasu *et al.*, 2000; Tudzynski *et al.*, 1998). In fungi, the GA biosynthesis genes are found on one chromosome, but in plants, they are found randomly on multiple chromosomes (Hedden *et al.*, 2001; Kawaide, 2006). Plants produce low amount of GA₃, therefore the GA₃ is produced for industrial purposes by microorganisms. Industrially the gibberellic acid can be produced by submerge fermentation, but this process presents low yield with high production costs and hence higher sale value, nevertheless other alternative process to reduce costs of GA₃ production in Solid-State-Fermentation (SSF) that allows the use of agro-industrial residues (Silva *et al.*, 2013).

One or two genes encode the enzymes responsible for the first steps of GA biosynthesis in *Arabidopsis* and rice (Yamaguchi, 2008). The null alleles of the genes encoding CPS, KS, and KO result in GA-deficient *Arabidopsis* dwarves (Koornneef and Veen, 1980). Multigene families encode the 20DDs that catalyze the formation of GA₁₂ to bioactive GA₄ (Yamaguchi, 2008).

AtGA3ox1 and AtGA3ox2, two of the four genes that encode GA3ox in *Arabidopsis*, affect vegetative development (Mitchum *et al.*, 2006). Environmental stimuli regulate AtGA3ox1 and AtGA3ox2 activity during seed germination (Yamauchi *et al.*, 1998; Yamauchi *et al.*, 2004). In *Arabidopsis*, GA20ox overexpression leads to an increase in GA concentration (Coles *et al.*, 1999; Huang *et al.*, 1998).

2.5.4 Cytokinins

Cytokinins are compounds with a structure that fits to adenine which promote cell division and have other similar functions to kinetin. Kinetin was

the first cytokinin discovered and so named because of the compounds ability to support cytokinesis (cell division). Though it is a natural compound, It is not made in plants, and is therefore considered a "synthetic" cytokinin (meaning that the hormone is synthesized somewhere other than in a plant). The most common form of naturally occurring cytokinin in plants today is called zeatin which was isolated from corn (*Zea mays*). Cytokinins have been found in almost all higher plants as well as mosses, fungi, bacteria, and also in tRNA of many prokaryotes and eukaryotes. Today there are more than 200 natural and synthetic cytokinins combined. Cytokinin concentrations are highest in meristematic regions and areas of continuous growth potential such as roots, young leaves, developing fruits, and seeds (Arteca, 1996; Mauseth, 1991; Raven, 1992; Salisbury and Ross, 1992).

2.5.4.1 History of Cytokinins

Haberlandt Gottlieb (1913) discovered that a compound found in phloem had the ability to stimulate cell division. Van Overbeek (1941) discovered that the milky endosperm from coconut had this ability. He also showed that various other plant species had compounds which stimulated cell division (Van Overbeek, 1941). Jablonski and Skoog (1954) extended the work of Haberlandt showing that vascular tissues included compounds which promote cell division. The first compound isolated that induced plant cytokinesis, and named kinetin, was derived from autoclaved herring sperm (Miller *et al.*, 1955). It promoted tobacco pith parenchyma differentiation in culture and stimulated totipotent plant cell growth (Sieberer *et al.*, 2003).

However, the first naturally occurring cytokinin was isolated from corn (*Zea Mais*) in 1961 by Miller and called zeatin. Letham reported that zeatin acts as a factor inducing cell division and later described its chemical properties (Letham, 1963). Since that time, many others naturally occurring cytokinins have been isolated and today there are more than 200 natural and synthetic cytokinins combined (Arteca, 1996; Salisbury and Ross, 1992).

2.5.4.2 Biosynthesis and Metabolism of Cytokinins

Cytokinin is generally detected in higher concentrations in meristematic regions and growing tissues. They are believed to be synthesized in the roots and translocated via the xylem to shoots. Cytokinin biosynthesis happens through the biochemical modification of adenine. The process by which they are synthesized is as follows (McGaw, 1995; Salisbury and Ross, 1992):

1. A product of the mevalonate pathway labeled isopentyl pyrophosphate is isomerized .
2. This isomer can then respond with adenosine monophosphate (AMP) with the aid of an enzyme called isopentenyl AMP synthase .
3. The result is isopentenyl adenosine-5'-phosphate (isopentenyl AMP).
4. This product can then be converted into isopentenyl adenosine by removal of the phosphate by a phosphatase and further converted to isopentenyl adenine by removal of the ribose group.
5. Isopentenyl adenine can be converted to the three major forms of naturally occurring cytokinins.
6. Other pathways or slight alterations of this one probably lead to the other forms.

Degradation of cytokinins occurs largely due to the enzyme cytokinin oxidase. This enzyme removes the side chain and releases adenine. Derivatives can also create but the pathways are more complex and poorly understood.

2.6 Bagging Fruit

During their growth and development, fruit undergo several physical and chemical changes and are susceptible to insect pest infestations, bird attack, various pathogens, and mechanical damage, all of which can reduce their commercial value and thereby cause significant yield and economic losses. To prevent the losses caused by biotic and abiotic factors, several good agricultural practices (GAP) are becoming popular throughout the World (Sharma, 2009).

Among several such alternatives, pre-harvest fruit bagging has emerged as an effective approach in different parts of the World. In this technique, individual fruit or fruit bunches are bagged on the tree for a specific period. Bagging is a physical protection technique, commonly applied to many fruit, which not only improves their visual quality by promoting peel coloration and reducing the incidence of fruit cracking and russeting, but can also change the micro-environment for fruit development (Fan and Mattheis, 1998), which can have multiple effects on internal fruit quality. Bagging has been used extensively in several fruit crops to improve skin color and to reduce the incidence of disease, insect pests, mechanical damage, sunburn of the skin, agrochemical residues on the fruit, and bird damage (Bentley and Viveros, 1992; Kitagawa *et al.*, 1992; Hofman *et al.*, 1997; Joyce *et al.*, 1997; Tyas *et al.*, 1998; Amarante *et al.*, 2002a; Xu *et al.*, 2010). Pre-harvest bagging of fruit is practiced in Japan, Australia, and China during peach, apple, pear, grape, and loquat cultivation in order to optimize fruit quality by reducing physiological and pathological disorders and to improve fruit coloration to increase market value (Joyce *et al.*, 1997) leading to an improved appearance (Amarante *et al.*, 2002b).

2.6.1 Effects of bagging on fruit size and weight

After fruit set, fruit growth slowly and increase in size to maturity. Covering fruit with a bag at a particular developmental stage may influence their growth and size. Reports on the effects of fruit bagging on fruit size and weight have been contradictory, and may be due to differences in the type of bag used, fruit age at bagging, fruit and cultivar responses, prevailing climatic conditions, and/or the storage conditions of fruit after harvest (Tyas *et al.*, 1998; Zhen *et al.*, 2000; Wang *et al.*, 2002; He *et al.*, 2003; Huang *et al.*, 2007; Chen *et al.*, 2012; Zhou *et al.*, 2012). Fruit bagging can increase, reduce, or have no effect on fruit weight and size.

Positive effects of bagging on fruit growth, size, and weight have been reported by Stover and Simmonds (1987), Xu *et al.* (2008), Watanawan *et al.*

(2008), Yang *et al.* (2009), Harhash and Al-Obeed (2010), Chonhenchob *et al.* (2011), and Zhou *et al.* (2012). For example, Xu *et al.* (2008) reported that bagging with plastic bags increased fruit weight in carambola when applied 10 d after full bloom. Watanawan *et al.* (2008) reported that bagging ‘Nam Dok Mai #4’ mango fruit with two-layer paper bags (black inside with brown, or brown and waxed, or white outside), newspaper, or golden paper bags and non-bagged fruit for 52 d increased fruit weight. Yang *et al.* (2009) reported that bagging promoted longan fruit development, resulting in larger-sized fruit. Similarly, Harhash and Al-Obeed (2010) reported that bunch bagging in ‘Succary’ and ‘Khalas’ date palm cultivars using blue bags increased finger and bunch weights significantly. Chonhenchob *et al.* (2011) studied the effects of pre-harvest bagging with different wave length selective bags on ‘Nam Dok Mai #4’ mango in Taiwan and reported that bagging increased fruit weight, size, and sphericity over un-bagged fruit. Individual fruit size in ‘Xiangtian’ olive was improved by Shengda bags in China (Zhou *et al.*, 2012).

2.6.2 Effects of bagging on fruit colour

Fruit colour is the fundamental that attracts consumers. An attractive colour improves the physical appearance of fruit, which helps to get better prices in domestic and export markets. Several studies have indicated that pre-harvest fruit bagging can promote or inhibit fruit colour development.

Bagging improves the colour of fruit by increasing their anthocyanin content. For example, in litchi, semi-transparent-CP bags resulted in excellent skin colouration in fruit (Tyas *et al.*, 1998, Chen and Li, 1999, Hu *et al.*, 2001). In earlier studies, fruit bagging had been reported to inhibit colour development in apple (Proctor and Loughheed, 1976); however, it has now been established that fruit bagging increases the light sensitivity of fruit and stimulates anthocyanin synthesis when the fruit are re-exposed to light after bag removal (Ju *et al.*, 1995a,b; Ju, 1998).

The fundamental effect of fruit bagging is to inhibit colour development, rather than to promote it (Proctor and Loughheed, 1976). The effect depends on the stage of fruit development at which it is bagged, the bagging date, the kind of bag used, the date of bag removal, and the climatic conditions of the area (Ju *et al.*, 1995a,b; Amarante *et al.*, 2002a). For example, Ju (1998) reported that bagging inhibited the accumulation of anthocyanins in the skin of “Delicious” apple fruit. However, when fruit were covered with three-layered bags and exposed to light, they started to accumulate anthocyanins rapidly, with a maximum after 3 days of light exposure. Kwan *et al.* (2000) reported that bagging “Yuzu” citrus (*Citrus junos* L.) fruit before early September with recycled Japanese phone-book paper (PBP) resulted in less colouration. Murray *et al.* (2005) reported that bagged plum fruit had a greener ground colour, and a poorer red colour than un-bagged fruit. Wei *et al.* (2006) reported that the content of anthocyanins was notably lower in bagged “Red Fuji” apples than in un-bagged control fruit; however, 5 days after bag removal, the anthocyanin content surpassed that in the controls. Takada *et al.* (2006) found that there was no significant difference in the rate of reddish-pulp development in “Takei Hakuho” peach between un-bagged fruit and those in white and orange double bags, but fruit covered with orange and black double bags showed a lower rate of reddish-pulp development at harvest.

2.6.3 Biotic factors influenced by fruit bagging

2.6.3.1 Pest control

Pre-harvest fruit bagging is a good technique to maintain a physical separation between the environment and the produce. One of the most significant effects of fruit bagging has been protection from the damage caused by insect pests. Several studies have indicated that bagging reduced the incidence of fruit fly in guava (Pereira, 1990; Marera-Montoya *et al.*, 2010), mango (Buganic *et al.*, 1997; Hofman *et al.*, 1997; Sarker *et al.*, 2009) codling moth in apple (Bentley and Viveros, 1992), woolly aphid in apple

(Teixeria *et al.*, 2011a,b), fruit borer in litchi (Debnath and Mitra, 2008), San Jose scale in apple (Sharma *et al.*, 2013), and fruit borer in pomegranate (Bagle, 2011). Thus, it appears that pre-harvest fruit bagging can be a useful approach to reduce the losses caused by insects, the control of which otherwise requires the application of several insecticides.

2.6.3.2 Disease control

Fruit bagging also prevents pathogens from reaching the developing fruit, which protects them from several diseases that can cause major losses. Bagging can protect some fruit from several pathogens, yet in other fruit it may not have any effect (Kitagawa *et al.*, 1992; Amarante *et al.*, 2002a; Senghor *et al.*, 2007; Chonhenchob *et al.*, 2011). Fruit bagging has been reported to reduce the incidence of anthracnose and stem-end rot in mango (Buganic *et al.*, 1997; Hofman *et al.*, 1997; Senghor *et al.*, 2007; Chonhenchob *et al.*, 2011), sooty blotch and fly speck in apple (Sharma *et al.*, 2013), and anthracnose and fruit rot in guava and loquat, respectively (Martins *et al.*, 2007; Ko *et al.*, 2010). In contrast, Fan and Mattheis (1998) reported that bagging during fruit development increased superficial scald and eliminated “stain” during cold storage. Similarly, bagging increased the incidence of brown spot in apple (Wang *et al.*, 2011). Thus, fruit bagging can be a beneficial practice for producing higher quality fruit, without or with less use of chemicals to control diseases.

2.6.4 Effect of bagging on fruit nutrient concentrations

Fruit contain several nutrients which contribute to their quality. Since fruit bagging is usually done in the orchard during the fruit development stage, it may influence the nutrient composition of fruit. For instance, Dong *et al.* (2007) reported that apple fruit covered with paper bags had the lowest concentration of calcium (Ca), but other bags increased this. Kim *et al.* (2008a) reported that bagging fruit in Ca-coated paper 4 - 5 weeks after full-bloom increased the Ca concentration in the skin of “Gamhong” apple in Korea. The concentrations of N and P in pear fruit were not significantly

affected by bagging, but the concentrations of K, Ca, Mg decreased by 9.6%, 38.9% and 6.7%, respectively (Lin, 2008). Similarly, Wang *et al.* (2010a) found that the Ca concentrations in bagged apple fruit were higher than in un-bagged. Accordingly, the incidence of bitter pit in bagged fruit was lower than in un-bagged fruit (Sharma *et al.*, 2013). In contrast, bagging of “Keitt” mango with white paper bags for up to 56 days reduced Ca concentrations, but prolonged bagging (82 – 131 days) did not have such an effect (Hofman *et al.*, 1997). Similarly, Beasley *et al.* (1999) found no significant differences in either skin or flesh Ca concentrations between bagged (plastic or paper) and un-bagged fruit. Amarante *et al.* (2002a) reported that pre-harvest bagging of “Doyenne du Comice” pear fruit with micro-perforated polyethylene bags, 30 DAFB, did not affect flesh concentrations of N, P, K, Ca, or Mg. Similarly, Wang *et al.* (2011) reported no significant effect of bagging on Ca concentrations in the skin of “Huangguan” pear fruit. Bagged “Fuji Suprema” apple fruit had higher Ca concentrations, which decreased the incidence of bitter pit during storage (Terxeria *et al.*, 2011b).

3. MATERIALS AND METHODS

3.1 Plant materials and experimental design

The experiment was conducted at the Tropical Fruit Orchard, Department of Plant Industry, National Pingtung University of Science and Technology, Taiwan from July 2014 to June 2015.

3.1.1. Experiment I: The effects of plant growth regulators (PGRs) on the productivity and quality of abiu fruit of autumn and winter

The experiment was carried in autumn and winter season 2014. The experiment consists of 4 treatments including control. Twenty uniformed branches were taken as an experimental unit. The selected uniformed branches were sprayed with GA₃ 10 ppm, NAA 10 ppm and Cytokinin 10 ppm and control. The design used in the experiment was Completely Randomized Design (CRD). Three growth regulators GA₃, NAA, and Cytokinin were applied during flowering.

3.1.2. Experiment II: Influence of Gibberellic acids (GA₃) at different concentrations on the productivity and quality of abiu fruit

The experiment was conducted in February 2015. Twenty uniformed branches were taken as an experimental unit. The experiment was conducted with four treatments namely (1) control, (2) 10 ppm, (3) 50 ppm, and (4) 100 ppm of GA₃. Different concentrations of GA₃ were sprayed during flowering. The design used in the experiment was Completely Randomized Design (CRD).

3.1.3. Experiment III: The effects of bagging on the productivity and quality of abiu fruit

The experiment was performed from December 2014 to March 2015. Twenty five uniformed branches were taken as an experimental unit. The

experiment consists of five treatment with different the kind of bag including a) the control, b) white paper bag, c) black plastic bag, d) white plastic bag, and e) blue paper bag. The design used in the experiment was Completely Randomized Design (CRD).

3.2 Bagging method

Fruit were bagged with four kinds: white paper bag, black plastic bag, white plastic bag and blue paper bag. Bagging fruit when young fruit period, 20 days after flowering. Fruit were bagged on 05 July; fruit were harvested at commercial maturity on 15 September, 72 days after bagged.

3.3 Measurement of physiological parameters

3.3.1 Fruit weight and seed weight

For the fruit weight, 5 fruits of each treatment from treated branches were brought to the laboratory and their fruit weight was measured by weight.

3.3.2 Fruit yield

Yield per treatment was recorded by weighing and counting the total number of fruits per branch at the time of harvesting.

3.3.3 Fruit size, seed size and flesh thickness

At harvesting, final fruit size (length and diameter), seed size (length and diameter) and flesh thickness was determined with a Vernier caliper.

3.3.4 Firmness

Using NO.5 probe, diameter about 0.5cm, by physical property analysis (Shimadzu, EZ-Test 500N). Before measure, set insert distance at 10mm. Measure 3 point and average every fruit. Unit is kg/cm².

3.4 Measurement of biochemical parameters

3.4.1 Total soluble solids

Using pocket refractometer to measure soluble solid of sample. Also sampling 3 region: top, central, and down, every point just need one chunk, and use juicer press for juice, to drip on specular side of refractometer and close cover. Rinse by RO water and clean for dry every time.

3.4.2 Titratable acidity acid

Taking 10g fresh into 250ml plastic beaker and add 100ml RO water, then smashed by homogenizing, to filter in a 100ml beaker by NO.1 Filter paper, take 5ml filtrate into specific plastic cup. Use 0.1N NaOH to titrate (Mettler Toledo, DL53) until pH 8.1, it is the final of titrate.

$$\text{Titrate acid content} = A \times F \times B \times \frac{100}{S} \times \frac{1}{25} \times 100$$

A: titer of 0.1N NaOH (ml)

F: 0.1N NaOH (1)

S: weight of sample (g)

B: organic acid content of 1ml 0.1 NaOH (malic acid: 0.0067, citric acid: 0.0064, tartaric acid: 0.0075)

3.4.3 Vitamin C content

Take 5g fresh into 250ml plastic beaker, add 50ml HPO_3 solution, fresh smashed by homogenizing, to filter in 100ml beaker by NO.1 Filter paper, take 5ml filtrate into Erlenmeyer Flasks, add 5ml HPO_3 solution to mix. Use Indophenol for titration until changed to pink (compared with standard solution), record the amount of titration.

$$\text{Vitamin C (mg/100g)} = \frac{S}{T} \times \frac{V1}{V2} \times \frac{1}{W} \times \frac{1}{K} \times 100$$

S: titer of Indophenol of sample

T: titer of Indophenol of standard Vitamin C

V1: amount of HPO_3 solution at first time (50ml)

V2: amount of HPO_3 solution at second time (5ml)

W: weight of sample (g)

K: 1

3.4.4 pH value

For pH measurement, a small portion of the homogenous juice was centrifuged at $4000 \times g$ for 10 mins and a small sample from the clear supernatant was analyzed for pH using Microprocessor pH meter (Hanna Instruments).

3.4.5 Skin color determination

The peel color of the fruits was determined using a Minolta colorimeter, CR-300. Parameters 'L' (lightness), 'a' (greenness to redness) and 'b' (blueness to yellowness) were determined at three different spots around the top, middle and end of 4 the fruits. The sensor was calibrated with a white tile prior to color measurement. All fruits were numbered for identification and marked equatorially on the peel surface. The changes in color of the same samples were measured every day during storage. Sample averages were calculated and the color was expressed in L^* , a^* , b^* Hunter parameters, using the following formula: $(L^* \times a^*) / b^*$. The chlorophyll content of leaf was measured by the methods described by Hendry and Price (1993).

3.5 Statistical analysis

The data obtained from the study were analyzed using SAS 9.1 statistical software. The least significantly difference was calculated following a significance F-test (at $P= 0.05$).

4. RESULTS AND DISCUSSION

4.1. Experiment I: The effect of PGRs on the productivity and quality of Abiu fruit between autumn and winter season

4.1.1 The effect of PGRs on characteristics of fruit and yield

GA₃, NAA, and Cytokinin have increased the fruit size, firmness and yield in both autumn and winter season. Moreover, figures in autumn season were higher than that in winter season.

In autumn season, there were no significant differences among all PGRs treatments. In Table 1, the highest significant fruit characteristics were observed in GA₃ treatment (494.45 g weight, 99 mm diameter, 98.68 mm length) followed by NAA and Cytokinin treatment. The lowest was recorded in the control treatment (326.73 g weight, 82.51 mm diameter, 83.98 mm length). The highest of firmness was 3.88 kg/cm² in NAA treatment. The greatest yield was obtained by GA₃ treatment (Table 1) at 5.44 kg, and the lowest yield recorded in the control treatment at 3.59 kg.

In winter season, all PGRs spray treatments had a positive effect on fruit characteristics. GA₃ has increased the fruit characteristics. The plant in which GA₃ was applied, produced maximum sized fruit (415.63 g weight, 88.49 mm diameter, and 91.06 mm length) with maximum firmness (3.29 kg/cm²) (Table 2). The increase yield was due the positive effect of PGRs applications to the increase in fruit size.

This result is in agreement growth regulators such as gibberellic acid (GA₃) and naphthalene acetic acid (NAA) significantly increased fruit weight and size of some date cultivars (Mohammed and Shabana, 1980) and of several other fruit types (Faust, 1989; Westwood, 1993; Davis, 2004) possibly by increasing cell size and/or cell numbers. Fruit weight and size are one of the critical quality characteristics that affect dates marketing since, larger size is generally preferred over small ones especially the fresh consumed cultivars as 'Barhee' (Botes and Zaid, 1999; Al-Qurashi and Awad, 2011).

4.1.2 The effect of PGRs on flesh thickness

The flesh thickness in autumn season in GA₃, NAA, and Cytokinin was higher than the flesh thickness in winter season (Figure 4). In autumn, GA₃ treatment was 44.46 mm, higher than other treatments. Followed by NAA, Cytokinin and lowest was control treatment at 31.11 mm.

Similarly, flesh thickness in winter with GA₃ treatment accounted for a significant proportion of about 38.38 mm which was higher than other figures. There was no significantly between GA₃ and NAA treatment. Control treatment was lowest flesh thickness at 31.48 mm.

4.1.3 The effect of PGRs on seed size

The fruits of control treatment produced highest number of seed/ fruit amounting to about 3.2 seeds per fruit in both autumn and winter season (Table 3). This was followed by the remaining three with a seed number of 1.8 (GA₃), 2 (NAA), and 2.6 (Cytokinin) in autumn. Seed number per fruit of control treatment was significantly higher than the GA₃, NAA, and Cytokinin treatment. The total seed weight per fruit in autumn was higher than in winter. Seed weight was almost 8.95 % higher in control treatment compared to GA₃, 7.53 % compared to NAA, and 2.98 % compared to Cytokinin. The highest (6.37 g) seed weight was recorded in control followed by Cytokinin, NAA, and GA₃ with a weight of 6.18, 5.89, and 5.80 g respectively.

4.1.4 The effect of PGRs on Total soluble solids (TSS) and pH value

Total soluble solids measurement is considered to be an important parameter of quality of Abiu fruits. As can be seen from Figure 8, the growth regulators treatments significantly increased the TSS compared to control in both autumn and winter season. In this case, spraying with GA₃ resulted in the highest TSS (13.43% Brix) in winter, whereas the control produced the lowest TSS (11.27% Brix). The pH also significantly affected by treatments (Figure 5). The lowest fruit pH value in fruit juice was recorded in control compared

to all treatment in both autumn and winter season. The highest content of pH was exhibited with GA₃ (6.52) in autumn.

In many reports, it was generalized that fruits were shown to have higher levels of soluble solids and sugar, but lower level of acid by GA₃ compared with the non treated fruits (Gelmese *et al.*, 2010). For instance, Graham and Ballesteros (2006) reported that GA₃ increased proteins, soluble carbohydrates, ascorbic acid, starch and carotene in the tomato. The effect of Cytokinin on TSS content of fruit has been positively addressed in a number of studies (Lewis *et al.*, 1996; Antognozzi *et al.*, 1996).

Thakur *et al.* (1996) increased pH value of fruit is a desirable quality and essential factor accounting for flavor when the whole plant was sprayed with GA₃. Thus, relationship between increased pH and increased sugar content is the desirable fruit quality parameter (Erdal *et al.*, 2007; Fontes *et al.*, 2000) to reduce the risk of microbial spoilage and requires moderate conditions for processing and enzyme inactivation.

4.1.5 The effect of PGRs on Titratable Acidity (TA)

The result of Figure 6 showed that application GA₃, NAA and Cytokinin had a significantly affected on TA, in which all the growth regulators decreased the acidity percentage. The control treatment produced the highest TA amount of 0.36% in autumn and 0.38% in winter, whereas the minimum TA (0.25 - 0.32 %) in autumn and (0.30 - 0.34 %) in winter was recorded with Cytokinin, NAA, and GA₃ application. As a result it seems that GA₃ application had a reducing effect on the acidity acid values drastically. GA₃ applications greatly decrease effect on the acidity value compared to the control treatment. The reduction in Titratable acidity observed seems be attributed to the conversion of the organic acids to sugar during fruit ripening by applying of plant growth regulators.

4.1.6 The effect of PGRs on Vitamin C

In this study, GA₃ application significantly affected the Vitamin C

content of Abiu fruits in autumn and winter season (Figure 7). The highest Vitamin C content (11.36 mg/100 g) was recorded with GA₃ in winter and (10 mg/100g) in autumn, followed by the fruits treated with NAA and Cytokinin, with a Vitamin C content of 8.86 mg/100g in autumn and 9.14 and 9.43 mg/100 g in winter, respectively. The lowest amount of Vitamin C (8.57 mg/100 g in autumn, 8.86 mg/100 g in winter) was recorded in the control fruit.

Xiao et al. (2005) also reported that the application of growth regulators significantly reduced the acidity and increased vitamin C contents of the citrus fruits.

Table 1. The effects of plant growth regulators on fruit characteristics of abiu fruit in autumn 2014

Treatment	Weight (g)	Diameter (mm)	Length (mm)	Firmness (kg/cm ²)	Yield (kg)
Control	326.73b	82.51b	83.98b	2.42b	3.59b
GA ₃ 10 ppm	494.45a	99.00a	98.68a	3.88a	5.44a
NAA 10 ppm	474.02a	96.23a	97.56a	4.07a	5.21a
Cytokinin 10 ppm	470.00a	96.28a	96.32a	3.76a	5.17a

(Different alphabets mark significant differences, $P < 0.05$ by LSD, $n=4$)

- PGRs were applied on 31/06/2014

Table 2. The effects of plant growth regulators on fruit characteristics of abiu fruit in winter 2014

Treatment	Weight (g)	Diameter (mm)	Length (mm)	Firmness (kg/cm ²)	Yield (kg)
Control	360.90b	85.46b	83.52b	2.57c	3.97b
GA ₃ 10 ppm	415.63a	88.49a	91.06a	3.29a	4.57a
NAA 10 ppm	368.04b	86.24a	85.01ab	2.87b	4.05b
Cytokinin 10 ppm	402.46a	86.03a	86.36ab	2.56c	4.43a

(Different alphabets mark significant differences, $P < 0.05$ by LSD, $n=4$)

- PGRs were applied on 26/09/2014

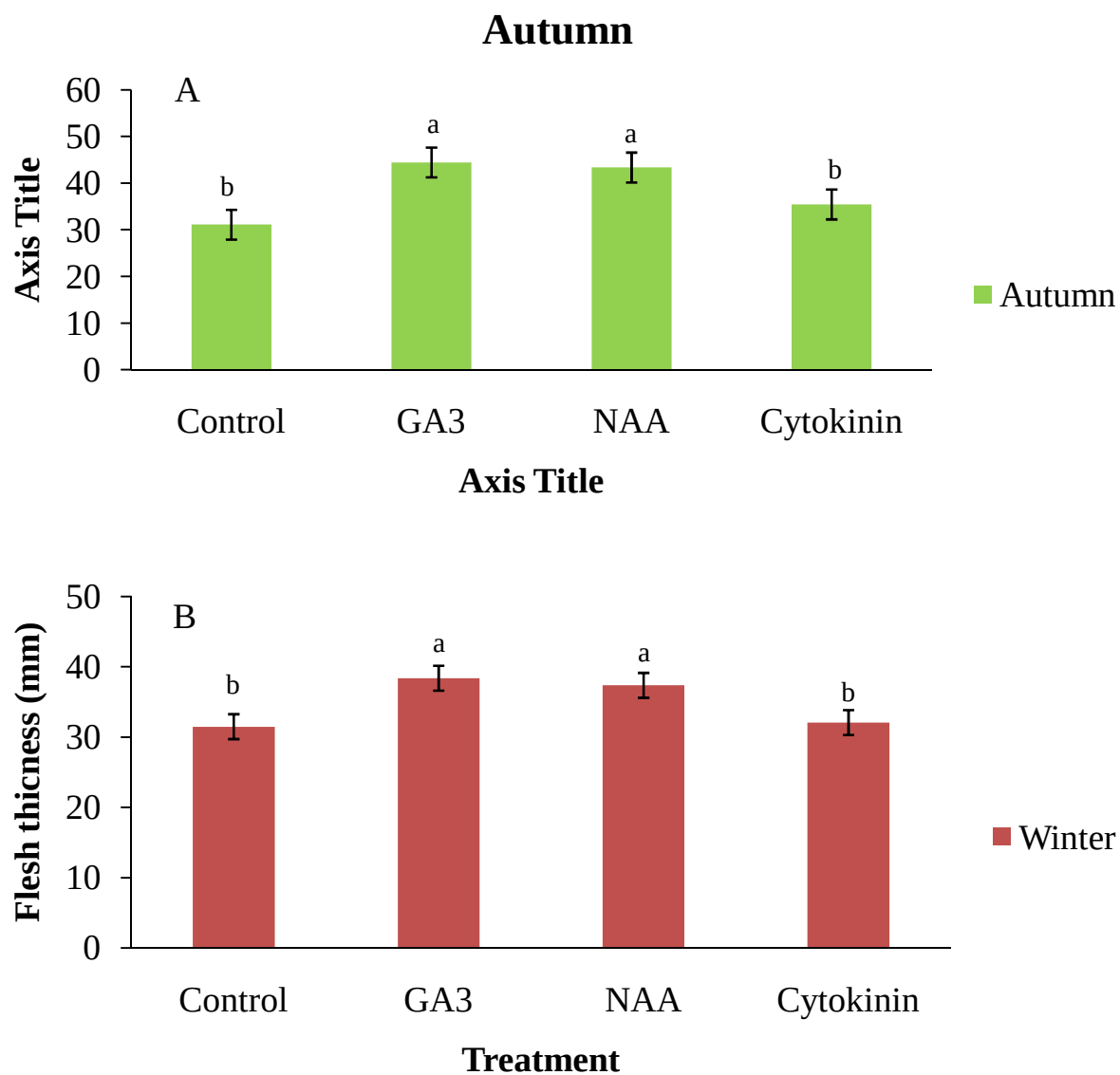


Figure 4. The effects of plant growth regulators on flesh thickness in autumn (A) and winter (B) (Different alphabets mark significant differences, $P < 0.05$ by LSD, $n=4$)

Table 3. The effects of plant growth regulators on seed size of Abiu fruit in autumn and winter season

Treatment	Autumn		Winter	
	No. of seed/ fruit	Seed weight/ fruit (g)	No. of seed/ fruit	Seed weight/ fruit (g)
Control	3.20c	6.37b	3.20c	6.22c
GA ₃	1.80a	5.80a	1.60a	5.71a
NAA	2.00a	5.89a	1.80a	6.03b
Cytokinin	2.60b	6.18a	2.40b	6.05b

(Different alphabets mark significant differences, $P < 0.05$ by LSD, $n=4$)

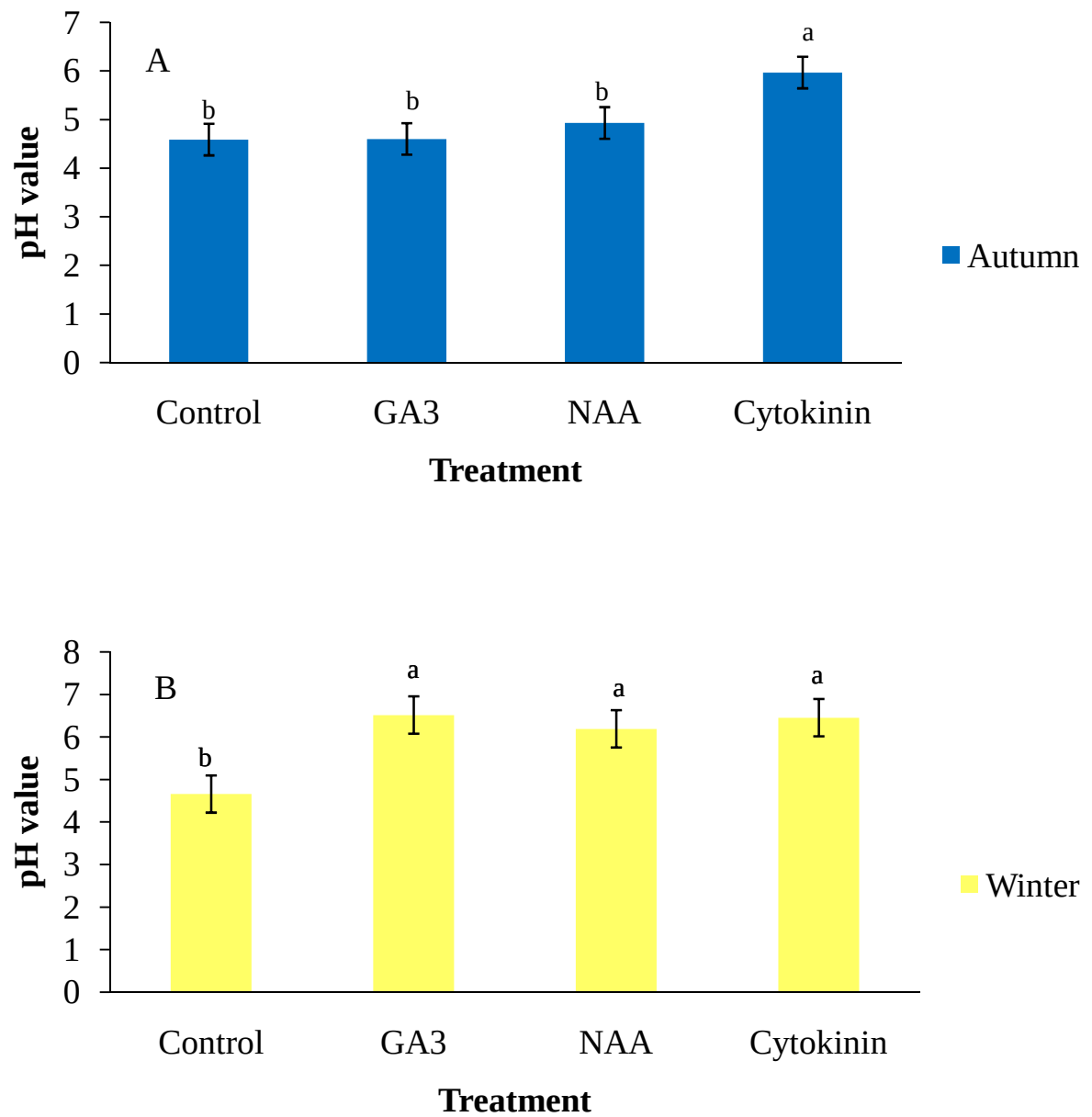


Figure 5. The effects of plant growth regulators on pH value in autumn (A) and winter (B) (Different alphabets mark significant differences, $P < 0.05$ by LSD, $n=4$)

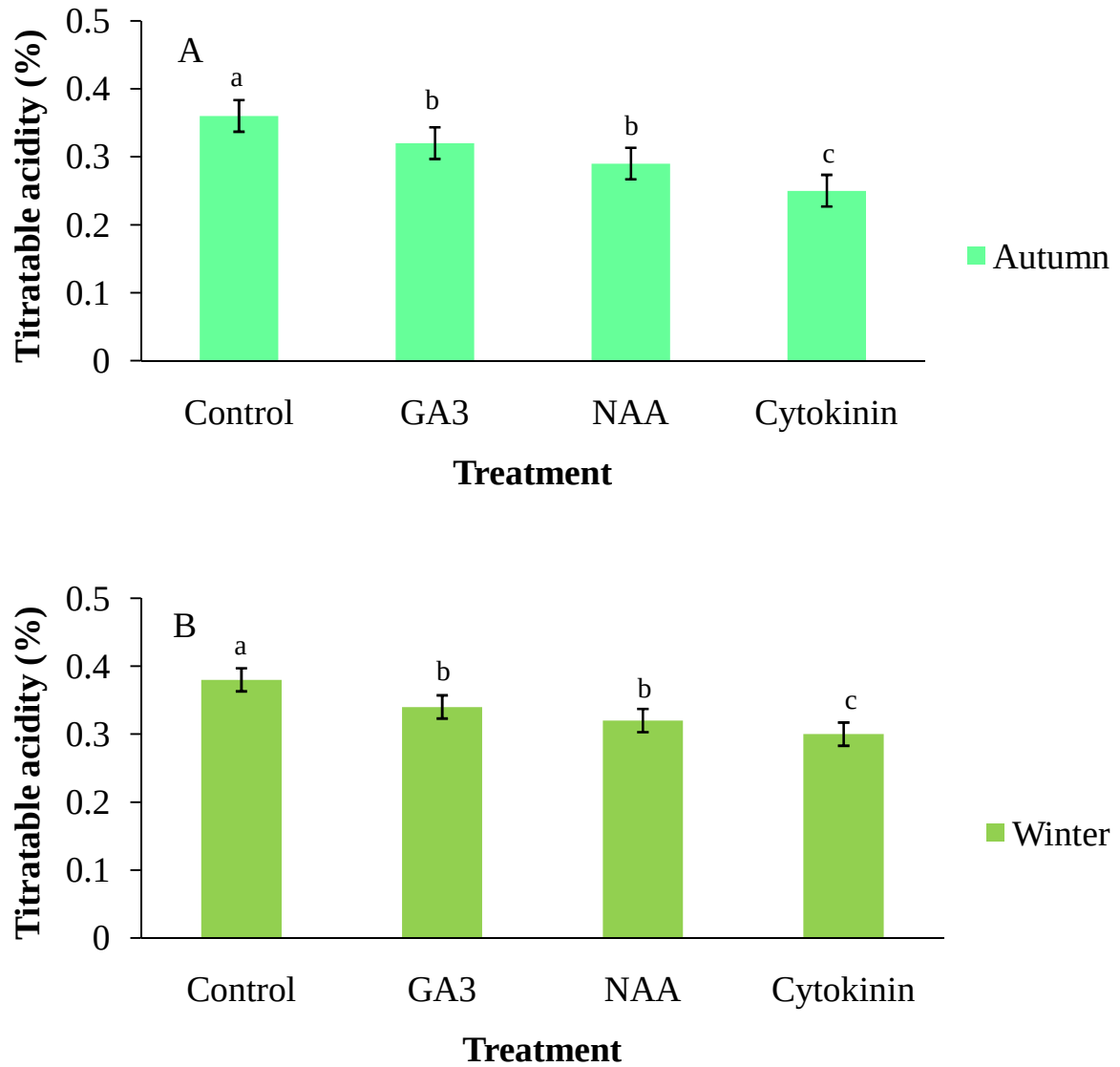


Figure 6. The effects of plant growth regulators on titratable acidity in autumn (A) and winter (B) (Different alphabets mark significant differences, $P < 0.05$ by LSD, $n=4$)

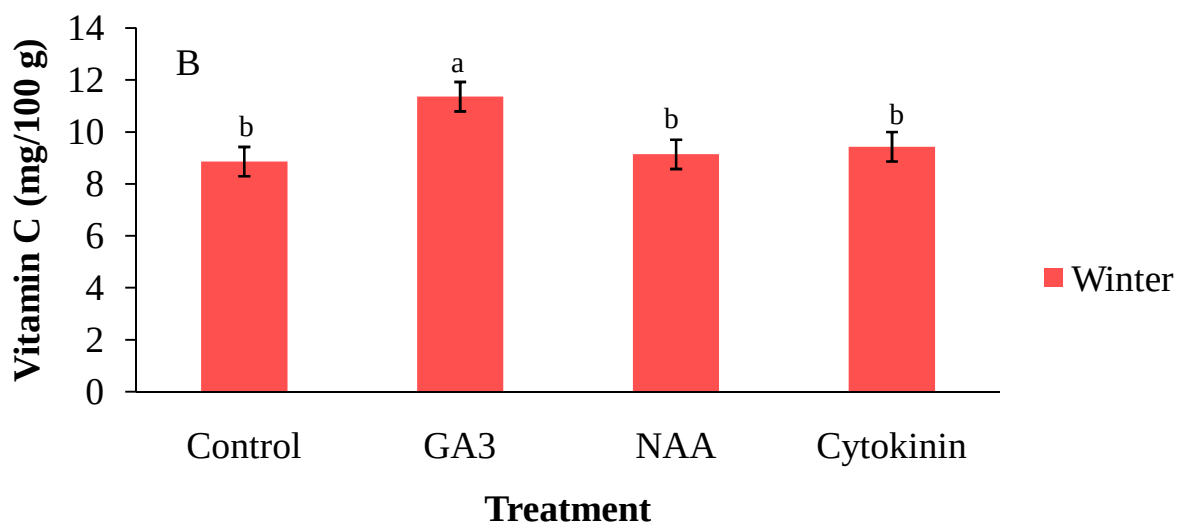
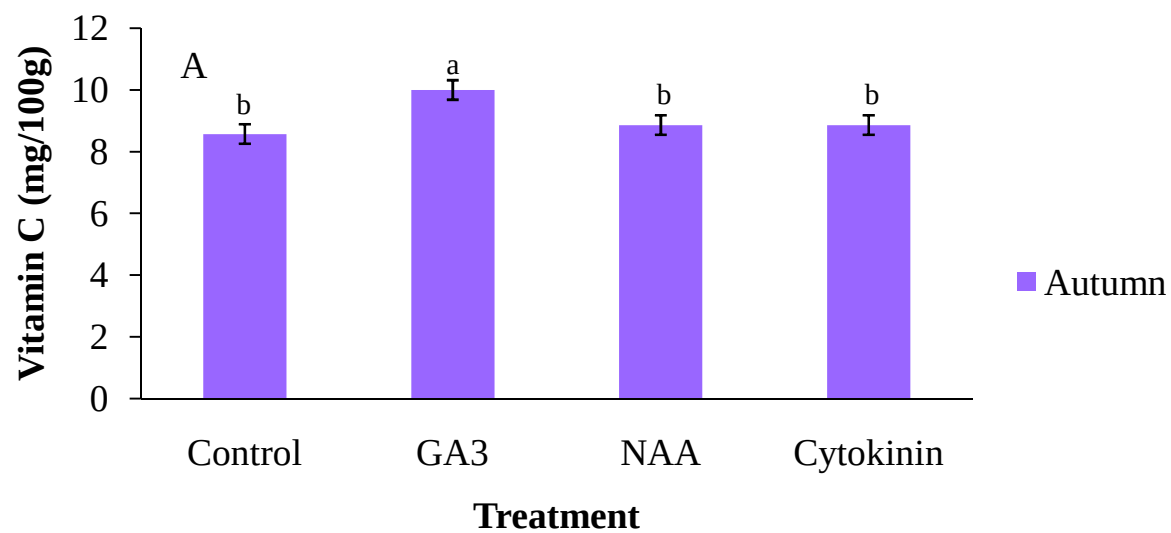


Figure 7. The effects of plant growth regulators on vitamin C in autumn (A) and winter (B) (Different alphabets mark significant differences, $P < 0.05$ by LSD, $n=4$)

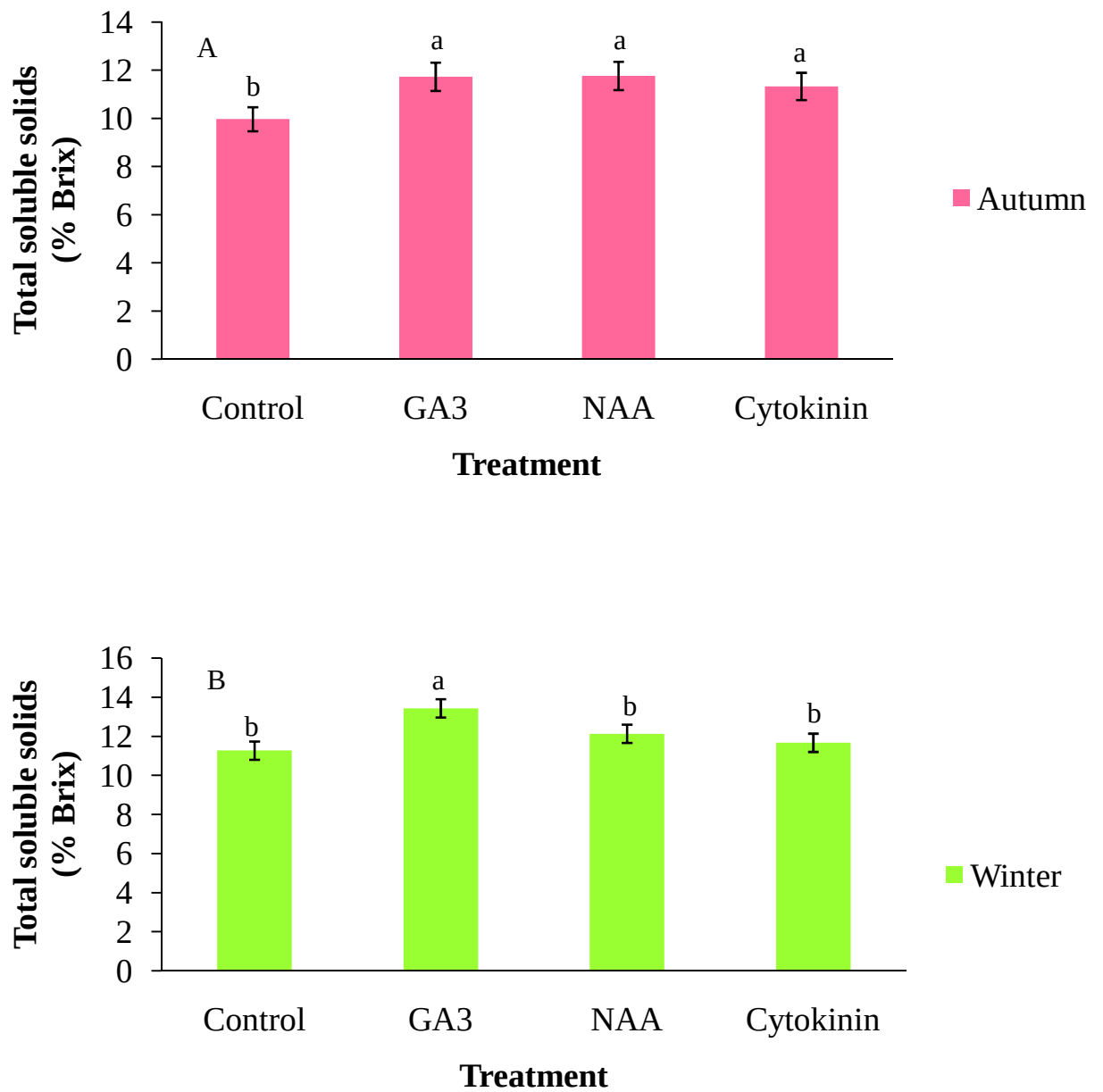


Figure 8. The effects of plant growth regulators on total soluble solids in autumn (A) and winter (B) (Different alphabets mark significant differences, $P < 0.05$ by LSD, $n=4$)



Control



10 ppm GA₃



10 ppm NAA



10 ppm Cytokinin

A

B

Figure 9. Effect of PGRs on abiu fruit in two seasons (A: autumn, B: winter)

4.2. Experiment II: Influence of Gibberellic acids (GA₃) at different concentrations on the productivity and quality of abiu fruit

4.2.1 The effect of GA₃ on fruit size, flesh thickness and yield

As shown in Table 4, the highest significant fruit weight was observed in GA₃ 50 ppm treatment (362.61 g) followed by control treatment at 326.73 g. The lowest weight was recorded in the GA₃ 100 ppm treatment at 280.23 g. According, yield of fruits was highest with GA₃ 50 ppm (3.99 kg) and lowest was GA₃ 100 ppm (3.08 kg). Several previous studies have reported that the GA₃ treatments could increase the fruit weight, peel thickness and fruit diameter of Valencia oranges (Almeida et al. 2004). Results for the fruit weight and yield were in agreement with that of Basak et al. (1998) who observed that GA₃ significantly influenced the fruit weight as well as the yield in cherry.

GA₃ 50 ppm and 100 ppm also increased fruit length. The plant was applied GA₃ 50 ppm produced maximum fruit length (92.26 mm) followed by GA₃ 100 ppm with 89.60 mm (Table 4). However, control treatment produced the highest fruit diameter (82.51 mm) followed by GA₃ 50 ppm. GA₃ spray at different rates significantly increased fruit weight, and improved both physical and biochemical quality characteristics of fruit especially at the higher rate. These results confirm those of Abou Aziz *et al.* (1982) and Hussein *et al.* (1993) for GA₃ application, or those of Hesami and Abdi (2010).

The flesh thickness (Table 4) was significantly influenced by the application GA₃; treatment with 50 ppm application produced the highest flesh thickness (42.18 mm) followed by 10 ppm and 100 ppm, and the lowest was control treatment (31.11 mm). Hence, GA₃ increased significantly flesh thickness after application compared to the untreated control.

4.2.2. The effect of GA₃ on Total soluble solids, Titratable acidity, and pH value

There were significant increases in TSS content recorded for GA₃ treatments when compared to the control treatment (Table 5). The highest TSS was recorded at 15.06 % Brix for GA₃ 100 ppm, followed by 14.69, 12.52 % Brix for GA₃ 50 and GA₃ 10 ppm. The lowest TSS was recorded at 9.94 % Brix for the control. The studies corroborated with the findings of Bal et al. (1984), Kale et al. (2000), Singh and Randhawa (2001) who reported beneficial effects of GA₃ in improving TSS of ber fruits. These findings were in agreement with the Basak et al. (1998) who reported that gibberellins could significantly increase the total soluble solids contents of the fruit in sweet cherry.

GA₃ decreased the Titratable acidity (TA), and the lowest significant TA was recorded at 0.19% for GA₃ 10 ppm. The highest TA recorded in the control treatment at 0.30%. There was no significant difference among control, GA₃ 50 and 100 ppm treatment (Table 5).

The result showed that GA₃ increased the pH values (Table 5). The highest pH value was recorded for GA₃ 50 ppm at 5.58, followed by GA₃ 100 ppm (5.24) and by GA₃ 10 ppm (5.19). The lowest pH value was observed in the control treatment at 4.47.

4.2.3 The effect of GA₃ on Vitamin C

In this study, GA₃ application significantly affected the vitamin C content. The highest vitamin C content (20.34 mg/100g) was recorded for GA₃ 10 ppm, followed by GA₃ 50 ppm with a vitamin C content of 19.37 mg/100 g. The lowest amount of vitamin C (8.85 mg/100g) was recorded in the control fruit (Figure 10). There was significant improvement in ascorbic acid content of ber fruits with GA₃ 50 ppm however, in association to present evaluations, GA₃ (10 and 25 ppm) when applied at slow growth phase exhibited the significant increase in ascorbic acid content of fruits (Pandey, 1999; Singh and Randhawa, 1976).

4.2.4 The effect of GA₃ on firmness

The firmness of Abiu fruits was significantly affected by GA₃ application. As a results showed that firmness increased with GA₃ application (Figure 11). The highest firmness was recorded in fruits treated with GA₃ 50 ppm at 4.22 kg/cm², whereas the firmness of untreated fruit was 3.82 kg/cm².

4.2.5 The effect of GA₃ on weight and length of seeds

The seed weight was significantly affected by GA₃ application. The results showed that seed weight decreased with GA₃ application (Figure 12). The highest seed weight was recorded in fruits treated with GA₃ 50 ppm at 3.81 g, followed by GA₃ 10 ppm with 4.19g. Whereas the seed weight of untreated fruit was 6.37 g.

GA₃ decreased seed length, and the lowest significant length of seed was recorded at 44.61 mm for control treatment. The highest seed length recorded in the GA₃ 50 ppm treatment at 35.59 mm. There was no significant difference between GA₃ 10 and 100 ppm treatment (Figure 12).

Table 4. The effects of GA₃ at different concentrations on fruit size, flesh thickness and yield of abiu fruit

Treatment	Fruit weight (g)	Fruit length (mm)	Fruit diameter (mm)	Flesh thickness (mm)	Yield (kg)
Control	326.73b	83.98c	82.51a	31.11b	3.59b
GA ₃ 10 ppm	320.58b	82.21c	77.33b	34.54b	3.53b
GA ₃ 50 ppm	362.61a	92.26a	81.13a	42.18a	3.99a
GA ₃ 100 ppm	280.23c	89.60b	79.79b	34.31b	3.08c

(Different alphabets mark significant differences, $P < 0.05$ by LSD, $n=4$)

- GA₃ were applied on 14/02/2015

Table 5. The effects of GA₃ at different concentrations on TSS, TA and pH of abiu fruit

Treatment	TSS (% Brix)	TA (%)	pH
Control	9.94c	0.30a	4.47b
GA ₃ 10 ppm	12.52b	0.19b	5.19ab
GA ₃ 50 ppm	14.69a	0.24a	5.58a
GA ₃ 100 ppm	15.06a	0.26a	5.24ab

(Different alphabets mark significant differences, $P < 0.05$ by LSD, $n=4$)

- GA₃ were applied on 14/02/2015

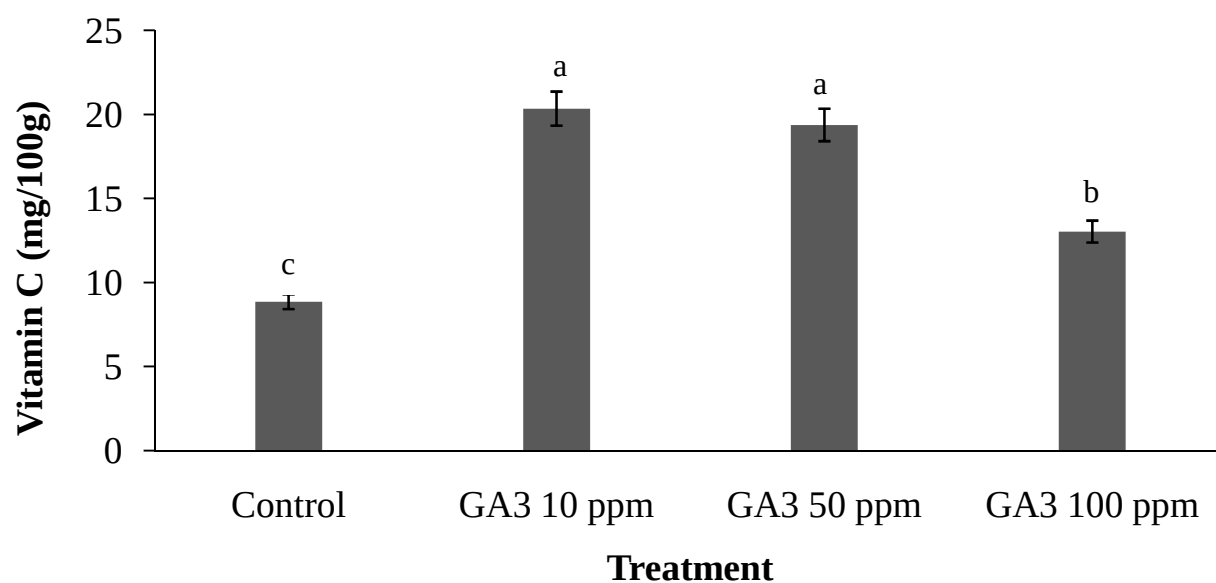


Figure 10. Effect of GA₃ at different concentrations on vitamin C of abiu
(Different alphabets mark significant differences, $P < 0.05$ by
LSD, $n=4$)

- GA₃ were applied on 14/02/2015

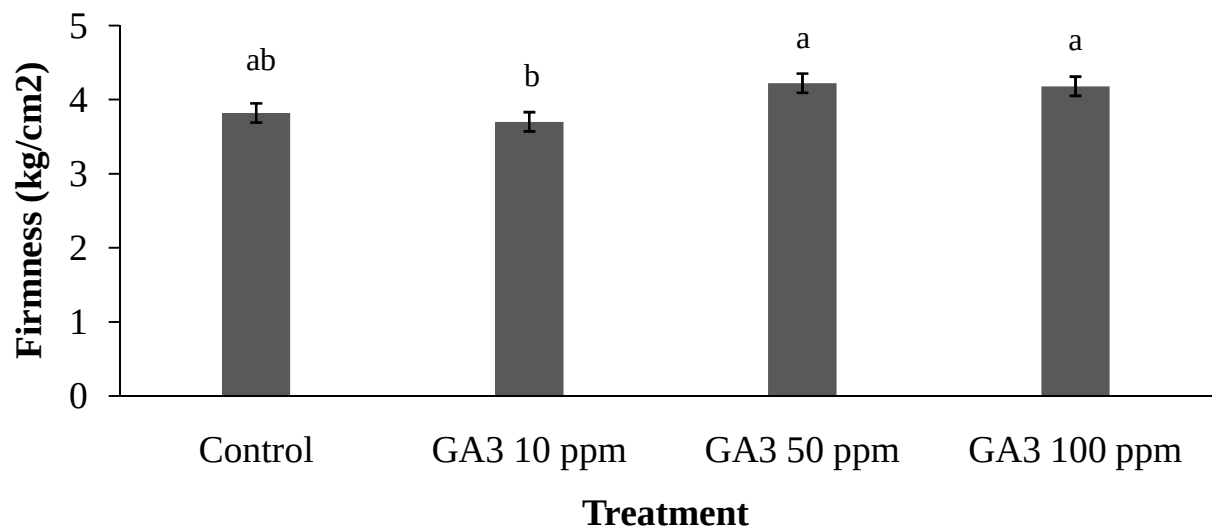


Figure 11. Effect of GA₃ at different concentrations on firmness of abiu
(Different alphabets mark significant differences, $P < 0.05$ by
LSD, $n=4$)

- GA₃ were applied on 14/02/2015

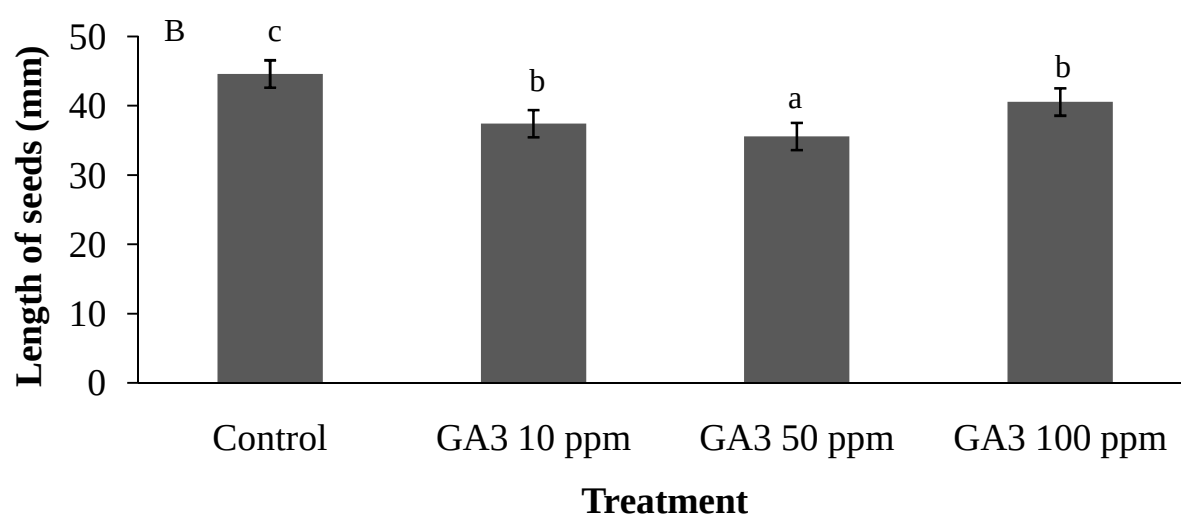
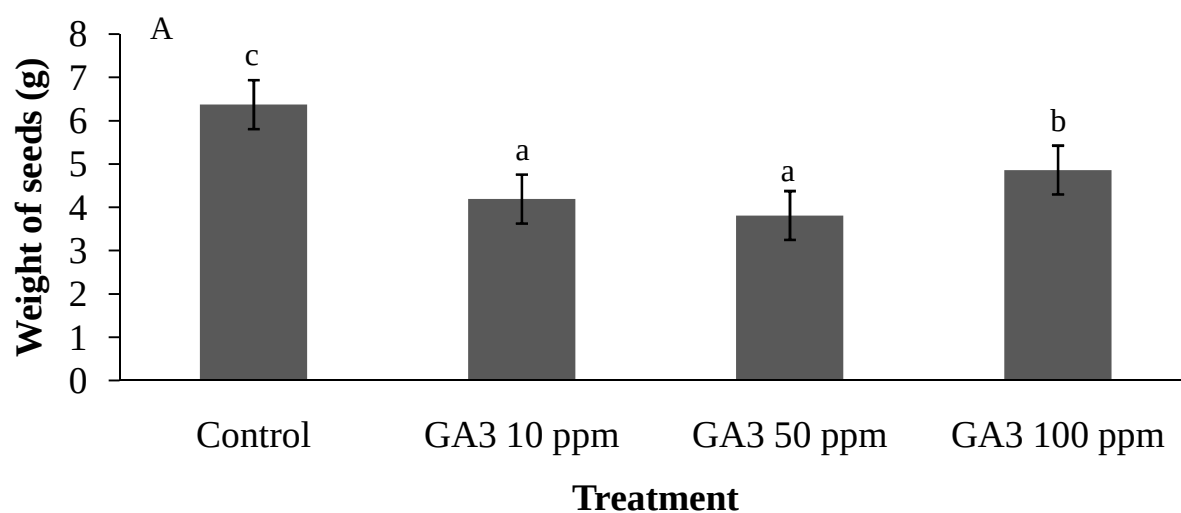


Figure 12. Effect of GA₃ on seed weight (A) and seed length (B) of abiu (Different alphabets mark significant differences, P < 0.05 by LSD, n=4)

- GA₃ were applied on 14/02/2015



Control



10 ppm GA₃



50 ppm GA₃



100 ppm GA₃

A

B

Figure 13. Effect of GA₃ at different concentrations on fruits (A) and seeds (B) of abiu

4.3. Experiment III: The effect of bagging on the productivity and quality of abiu fruit

4.3.1 The effect of bagging on fruit size, flesh thickness and yield

Bagging affects the size and weight of apple, pomegranate and banana (Arakawa *et al.*, 1994; Hasan *et al.*, 2001). All bagging treatments had a positive effect on fruit weight. As shown in Table 6, the highest significant fruit weight was observed in white paper bag treatment (477.13 g) followed by black plastic bag treatment at 445.46 g. The lowest weight was recorded in the control treatment at 326.73 g. Hence, bagging treatment substantially increased fruit weight when compared to the control treatment. Amarante *et al.* (2002a) also reported that pre-harvest bagging of ‘Doyenne du Comice’ pears had no effect on fruit size, weight

White paper bag and black plastic bag also increased fruit length and diameter. The plant was applied with white paper bag produced maximum sized fruit (97.77 mm diameter and 101.46 mm length) followed by black plastic bag with 97.17 mm diameter and 98.65 mm fruit length (Table 6). However, control treatment produced the lowest fruit diameter (82.51 mm) and minimum fruit length (83.98 mm).

The flesh thickness (Table 6) was significantly influenced by the application bagging; treatment with white paper bag application produced the highest flesh thickness (39.66 mm) followed by black plastic bag with flesh thickness of 37.3 mm, and the lowest was control treatment (31.11 mm). Hence, bagging increased significantly flesh thickness after application compared to the untreated control.

4.3.2 The effect of bagging on fruit color

This color change in flesh agreed to the previous reports (Lee, 2000; Lime *et al.*, 1954; Oberbacher *et al.*, 1960; Porras *et al.*, 1996; Ting *et al.*, 1980). For the L value of fruit color, the blue paper bag showed the lowest

value at 56.19, while the white paper bag had the highest value at 61.67 (Table 7).

For a value (red), control treatment showed the lowest value while the black plastic bag has highest value. There was no significance among control, white plastic bag and blue paper bag.

For the b value (yellow), it clearly that the highest value was found in white plastic bag, black plastic bag has lowest value (Table 7). It indicates that black plastic bag is poor for fruit lightness and also for the improvement of red and yellow components in the fruit color.

The control fruit had a higher chroma (C) than the bagged fruit indicating the higher red color saturation. Hue angle value has not changed so much in bagging treatment and control (Table 7).

Amarante et al. (2002a) reported that pre-harvest bagging of pears improved skin finish, resulting in a fruit with a more attractive light green colour, without reducing blush on the exposed side of the skin. The pears bagged until harvest had a yellow tinge, with high lightness value and hue angle (Huang *et al.*, 2009). Jia et al. (2005) stated that non-bagged peach fruit had a higher Chroma than the bagged fruit.

4.3.3 The effect of bagging on pH value, vitamin C and Titratable acidity

The result showed that bagging increased the pH values (Table 8). The highest pH value was recorded for white plastic bag at 6.47, followed by blue paper bag (6.45) and by white paper bag (5.70). However, there was no significant difference among white plastic bag, blue paper bag, and white paper bag treatments. The lowest pH value was observed in the control treatment at 4.58

In this study, bagging application significantly affected the vitamin C content. The highest vitamin C content (17.42 mg/100g) was recorded for white paper bag, followed by the white plastic bag and black plastic bag with

a vitamin C content of 13.40 mg/100 g. The lowest amount of vitamin C (9.14 mg/100g) was recorded in the control fruit and blue paper bag.

Bagging increased the titratable acidity (TA), and the lowest significant TA was recorded at 0.26% for white plastic bag. The highest TA recorded in the control treatment at 0.46%. There was no significant difference among white paper bag, white plastic bag, black plastic bag and blue paper bag treatments (Table 8). Kim et al. (2000b) showed that titratable acidity tended to increase by bagging with yellow paper of low light transmittance

4.3.4 The effect of bagging on firmness

The firmness of abiu fruits was significantly affected by NAA application. Our results showed that firmness increased with bagging application (Figure 14). The highest firmness was recorded in fruits treated with white paper bag at 3.15 kg/cm², whereas the firmness of untreated fruit was 2.42 kg/cm².

Bentley and Viveros (1992) reported that fruit firmness of “Granny Smith” apples was improved by brown paper bags when done at golf-size of fruit development. However, Hofman et al (1997) reported that fruit firmness was not affected by white paper bag in mango

4.3.5 The effect of bagging on total soluble solids

There were significant increases in TSS content recorded for bagging treatments when compared to the control treatment (Figure 15). The highest TSS was recorded at 11.7% Brix for white plastic bag, followed by 11.4, 11.37, 11.27% Brix for black plastic bag, white paper bag, blue paper bag, respectively. The lowest TSS was recorded at 9.97% Brix for the control. Hussein et al (1994) reported that bagging with paper bags significantly increased TSS content. Padmavathamma and Hulamani (1996) found that the total sugars varied significantly with bag colour.

Table 6. Effect of bagging on yield contributing characteristics in abiu fruit

Treatment	Fruit weight (g)	Fruit diameter (mm)	Fruit length (mm)	Flesh thickness (mm)	Yield (kg)
Control	326.73b	82.51b	83.98c	31.11c	3.59c
White paper bag	477.13a	97.77a	101.46a	39.66a	5.25a
Black plastic bag	445.46a	97.17a	98.65a	37.30a	4.90a
White plastic bag	390.42b	96.12a	96.4ab	36.65ab	4.29b
Blue paper bag	392.24ab	87.46ab	88.26bc	32.69b	4.31b

(Different alphabets mark significant differences, $P < 0.05$ by LSD, $n=4$)

- Bagging applied from 05/07/2014 – 24/09/2014

Table 7. Effect of bagging on fruit color of abiu

Treatment	L	a	b	C
Control	57.6b	-5.03b	35.2a	99.0a
White paper bag	61.6a	-0.68a	33.1a	91.0b
Black plastic bag	56.3b	-0.57a	32.8b	93.0b
White plastic bag	60.3a	-4.00b	36.8a	96.9a
Blue paper bag	56.1b	-3.36b	34.3a	98.2a

L = lightness, a = bluish-green/red-purple, b = yellow/blue, chroma (C) = $(a^2+b^2)^{1/2}$

(Different alphabets mark significant differences, $P < 0.05$ by LSD, $n=4$)

- Bagging applied from 05/07/2014 – 24/09/2014

Table 8. Effect of bag treatments on pH, vitamin C, and titratable acidity in abiu fruit

Treatment	pH	Vitamin C (mg/100 g)	Titratable acidity (%)
Control	4.59b	9.14b	0.25b
White paper bag	5.70a	17.42a	0.45a
Black plastic bag	4.66b	13.4a	0.46a
White plastic bag	6.48a	13.4a	0.26b
Blue paper bag	6.45a	9.14b	0.3ab

(Different alphabets mark significant differences, $P < 0.05$ by LSD, $n=4$).

- Bagging applied from 05/07/2014 – 24/09/2014

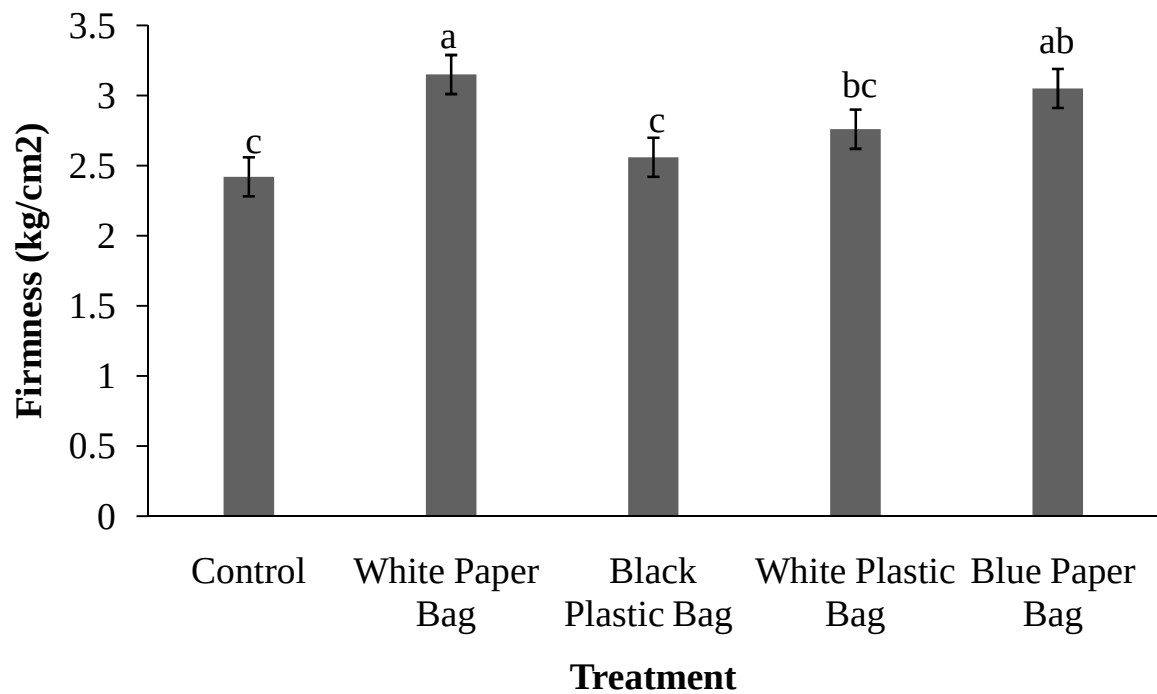


Figure 14. Effect of bagging on fruit firmness of bag (Different alphabets mark significant differences, $P < 0.05$ by LSD). S.E. $\pm$ (n=4)

- Bagging applied from 05/07/2014 – 24/09/2014

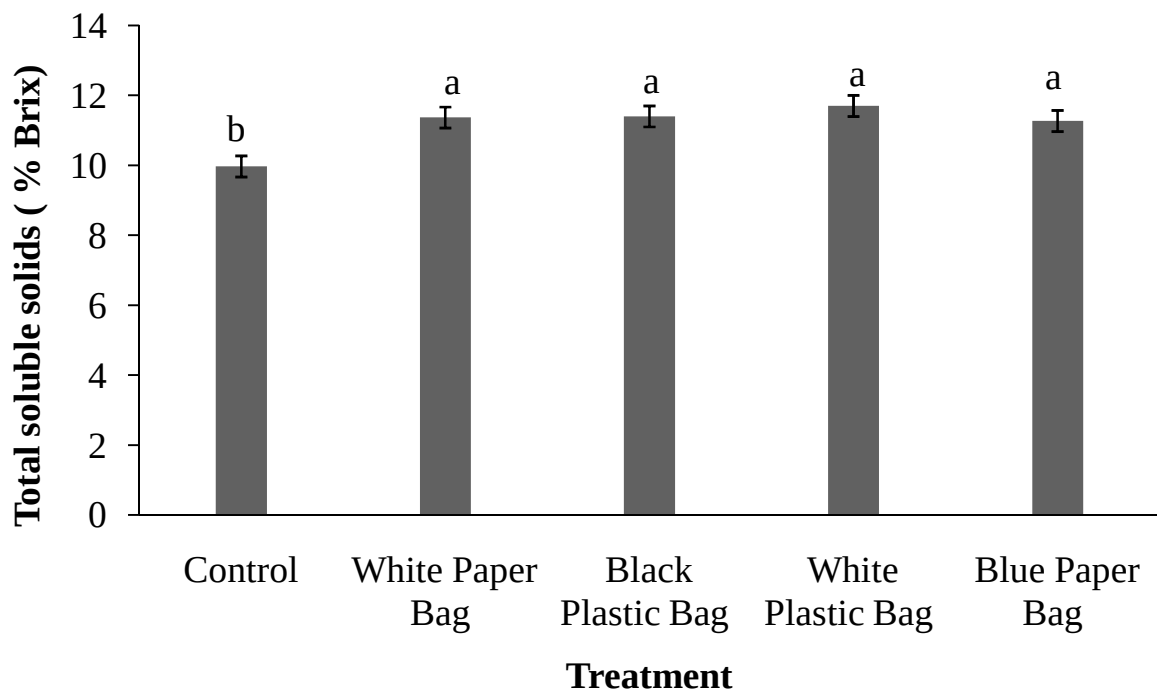


Figure 15. Effect of bagging on total soluble solids of abiu fruit (Different alphabets mark significant differences, $P < 0.05$ by LSD). S.E. $\pm$ (n=4)

- Bagging applied from 05/07/2014 – 24/09/2014



Figure 16. Four kinds of bag

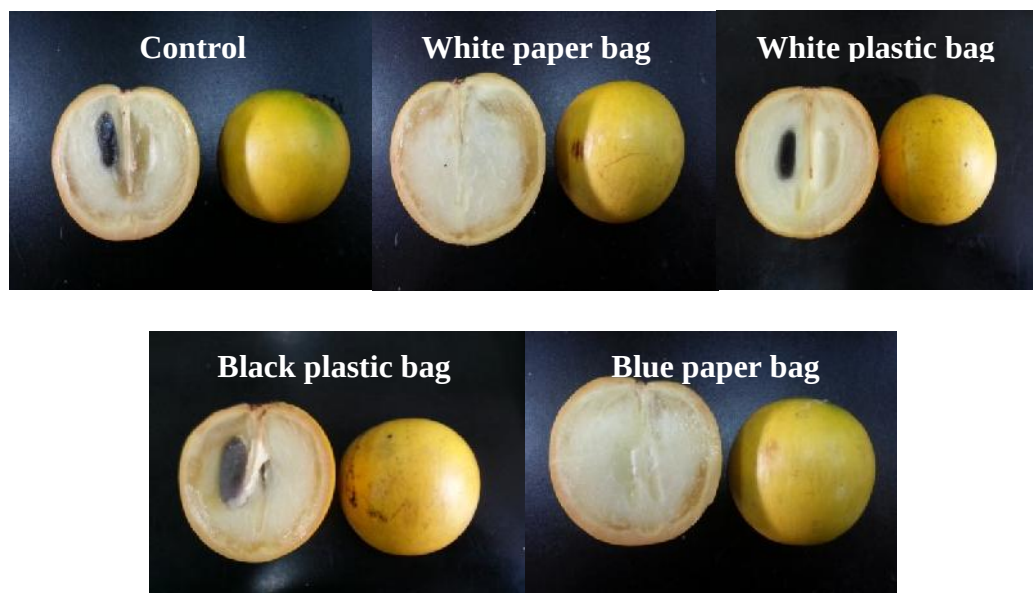


Figure 17. Effect of bagging on color of abiu fruit

5. CONCLUSIONS

Experiment 1: PGRs spray treatments had a positive effect on fruit characteristics and yield in both autumn and winter season. In Tables 1 - 2, the highest significant fruit size, firmness and fruit yield was GA₃ 10 ppm treatment and the lowest was recorded in the control treatment. However, figures of fruit characteristics and yield in autumn are higher than winter season.

Number of seeds and seed weight were increased by using different PGRs in both seasons (Table 3). The effect of GA₃ 10 ppm on number of seeds and seed weight was the highest when compared to other treatment.

The highest flesh thickness (Figure 4) was recorded in GA₃ treatment and the lowest flesh thickness was control treatment. In winter season, the flesh thickness was lower than in autumn season.

Results in Figures 5 & 8, indicated that there were significant increases in pH value and TSS content recorded in PGRs treatments when compared to the control treatment. The lowest pH and TSS were control treatment. In autumn season, both pH and TSS were lower than that in winter season.

Result in Figure 6 showed that PGRs spray treatments decreased the Titratable acidity (TA). The lowest significant TA was recorded in cytokinin treatment and the highest TA recorded in the control treatment.

In Figure 7, the highest vitamin C content was observed in the GA₃ 10 ppm and the lowest vitamin C content was recorded at the control treatment. The vitamin C content in autumn season was higher than that in winter season.

From the previous conclusion, the study suggested the best Plant Growth Regulators spraying treatment for Abiu grown under field condition was GA₃ 10 ppm in both autumn and winter season.

Experiment 2: GA₃ treatments (10, 50 and 100 ppm) responded well in improving in fruit characteristics, flesh thickness and yield (Table 4). The highest effect showed on GA₃ at 50 ppm. Moreover, fruit diameter with control treatment was higher than GA₃ treatments.

In Table 5, total soluble solids (TSS) and pH were affected significantly by GA₃ treatment at different concentrations. GA₃ 100 ppm was highest TSS when compared with other treatments, while GA₃ 50 ppm was highest pH value. Moreover, GA₃ was decreased the Titrable Acidity (TA), and control treatment was higher than other treatment.

GA₃ 50 ppm had a positive effect on firmness (Figure 11) and seed size (Figure 12). The highest of firmness was GA₃ 50 ppm, followed by GA₃ 100 ppm. Plants which applied GA₃ 50 ppm were highest of seed size (seed weight and seed length), and followed by GA₃ 10 ppm. The highest vitamin C content was observed in GA₃ 10 ppm and the lowest vitamin C content was recorded at the control treatment (Figure 10).

From the results, it can be concluded that GA₃ treatments had a positive effect on abiu grow, and GA₃ at 50 ppm was the best effect on yield and quality of fruit under field condition.

Experiment 3: As the results of bagging treatment showed that, fruit size (weight, length, and diameter), flesh thickness, yield (Table 6), and firmness (Figure 14) were the highest for white paper bag treatment compared to other treatments. These figures of bagging treatments were higher than un-bagged.

pH value was also highest in white plastic bag, while vitamin C (Table 8) was highest in white paper bag . Titratable acidity (TA) was increased in other treatments compared to control treatment (Table 8), and black plastic bag was highest

Total soluble solids (TSS) were higher in bagging treatment than other control treatment (Figure 15). Bagging had no significant effect on color of Abiu fruits (Table 7).

Hence, the best bagging treatment for abiu grown under field condition was white paper bag followed by black plastic bag, and white plastic bag.

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